

STDA- A1 - Spatial Analysis of Global Seismicity (2000–2025)

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Abstract—This report presents a rigorous spatiotemporal analysis of global seismic events (2000–2025) using a dataset of 175,947 records. The investigation is structured into four sequential tasks: (1) advanced preprocessing and feature engineering, (2) spatial autocorrelation analysis, (3) heterogeneity and stationarity diagnostics, and (4) predictive spatial regression modeling. By engineering a 47-feature dataset that accounts for cyclic temporality and outlier detection, we demonstrate that earthquake magnitudes exhibit significant positive spatial autocorrelation ($I \approx 0.51$, $p < 0.005$) with a characteristic range of 1,000–1,500 km. Theoretical diagnostics reveal extreme forms of spatial heterogeneity, where 80% of seismic energy is released in less than 20% of the domain (Lorenz Analysis), and stationarity tests confirm that statistical moments (mean, variance) are unstable across latitude and longitude. These findings necessitate the rejection of global OLS models in favor of geographically aware algorithms. Validation of Spatial Random Forest and Trend Surface models confirms their superior ability to predict magnitudes at unseen locations by effectively capturing local tectonic dependencies, thereby providing a robust framework for global seismic risk assessment.

Index Terms—Seismic events analysis, Spatiotemporal preprocessing, Feature engineering, Spatial autocorrelation, Spatial heterogeneity, Spatial regression

I. TASK 1 – DATA PREPROCESSING AND FEATURE ENGINEERING

This section details the comprehensive preprocessing pipeline applied to the USGS earthquake dataset [1] spanning from 2000 to 2025. The objective of this task is to transform the raw seismological data into a clean, feature-rich dataset suitable for downstream spatial analysis, autocorrelation studies, and regression modelling. The preprocessing encompasses data cleaning, advanced feature engineering, transformation of skewed distributions, outlier detection, dimensionality reduction, spatial clustering, and the computation of spatial lag features for autocorrelation analysis.

A. Data Loading and Initial Cleaning

The raw dataset was sourced from the United States Geological Survey (USGS) Earthquake Hazards Program and loaded as a comma-separated values (CSV) file containing 175,947 earthquake event records [1]. Each record includes attributes such as event time, geographic coordinates (latitude and longitude), depth, magnitude, magnitude type, station count (`nst`), azimuthal gap (`gap`), root-mean-square travel time residual

(`rms`), contributing network (`net`), event identifier, and place description.

The following initial cleaning steps were performed:

- **Column removal:** Columns with excessive missing values or limited analytical relevance were dropped, including `dmin`, `horizontalError`, `depthError`, `magError`, `magNst`, `locationSource`, and `magSource`.
- **Missing value filtering:** Rows with missing values in the critical columns — `mag`, `depth`, `latitude`, and `longitude` — were removed to ensure completeness of the core spatial and seismological attributes.
- **Temporal parsing:** The `time` column was converted to a `datetime` object to enable temporal feature extraction.

After cleaning, the dataset retained **175,947 records** with **15 attributes**, providing a robust foundation for subsequent analysis.

B. Temporal Feature Engineering

Temporal patterns are critical in seismological analysis, as earthquake occurrence rates may exhibit diurnal, seasonal, or long-term trends. The following temporal features were extracted from the `time` column:

- **Basic components:** `year`, `month`, `day`, `hour`, and `day_of_week`.
- **Cyclical encoding:** Standard integer representations of month and hour fail to capture the cyclical nature of time — for instance, December (month 12) is temporally adjacent to January (month 1). To address this, sine and cosine transformations were applied:

$$m_{\sin} = \sin\left(\frac{2\pi \cdot m}{12}\right), \quad m_{\cos} = \cos\left(\frac{2\pi \cdot m}{12}\right) \quad (1)$$

$$h_{\sin} = \sin\left(\frac{2\pi \cdot h}{24}\right), \quad h_{\cos} = \cos\left(\frac{2\pi \cdot h}{24}\right) \quad (2)$$

where m denotes the month (1–12) and h denotes the hour (0–23).

This encoding ensures that cyclically adjacent time periods are represented by numerically close feature values, which benefits distance-based and regression models.

C. Spatial Clustering via K-Means

To identify major seismically active zones, K-Means clustering ($K = 20$) was applied to the geographic coordinates (latitude, longitude) [3]. This groups earthquake events into 20 spatial clusters that approximate tectonic plate boundaries and other seismically active regions.

For each event, the **Euclidean distance to its assigned cluster centre** was computed as an additional feature (`dist_to_cluster_center`). Events located far from their cluster centre may represent spatially isolated seismic activity or transitional boundary zones.

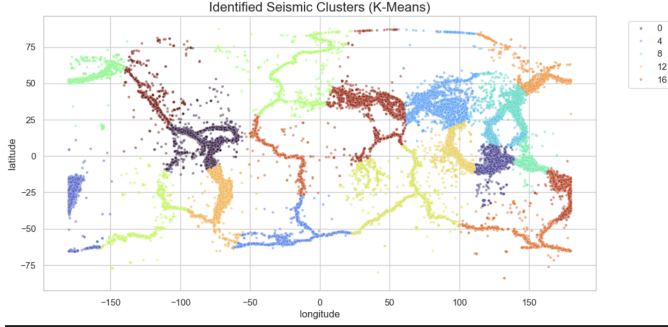


Fig. 1. Global distribution of earthquake events colour-coded by K-Means cluster assignment ($K = 20$). The identified clusters correspond closely to known tectonic plate boundaries, including the Pacific Ring of Fire, the Mid-Atlantic Ridge, and the Alpine-Himalayan Belt.

Figure 1 illustrates the spatial distribution of the 20 identified clusters. The clustering successfully delineates major seismogenic zones, with dense clusters along subduction zones (e.g., the western Pacific margin) and divergent boundaries (e.g., the Mid-Atlantic Ridge).

D. Spatio-Temporal Lag Features

To capture temporal dependencies within spatially proximate regions, lag features were computed within each K-Means cluster:

- **Hours since last event** (`hours_since_last_event`): The elapsed time in hours between consecutive events within the same cluster, sorted chronologically.
- **Previous magnitude** (`prev_mag`): The magnitude of the immediately preceding event within the same cluster.
- **Rolling mean magnitude** (`rolling_mag_5`): A 5-event rolling average of magnitude within the cluster, smoothing short-term fluctuations.

These features encode the local seismic history and are essential for capturing aftershock sequences and stress-transfer dynamics.

E. Distribution Transformation

Several raw features exhibit significant skewness, which can adversely affect the performance of parametric statistical methods and machine learning models. The skewness values of the original distributions were:

- gap: 0.424
- rms: 52.140
- depth: 3.407

The following transformations were applied:

- 1) **Logarithmic transformation** for depth: A shifted log transformation ($\log(1 + \text{depth} + |\min(\text{depth})| + 1)$) was applied to handle the heavily right-skewed depth distribution, including negative or zero depth values (shallow events).
- 2) **Yeo-Johnson Power Transformation** for gap and rms: The Yeo-Johnson method generalises the Box-Cox transformation to handle zero and negative values, producing approximately Gaussian distributions.

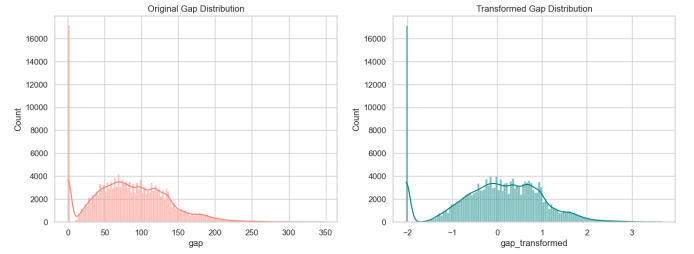


Fig. 2. Comparison of the azimuthal gap (`gap`) distribution before (left, salmon) and after (right, teal) Yeo-Johnson power transformation. The transformed distribution exhibits substantially reduced skewness and approximates a Gaussian shape.

As shown in Figure 2, the power transformation effectively normalises the gap distribution, reducing skewness and enabling more reliable statistical inference.

F. Outlier Detection Using Isolation Forest

Outlier detection is essential in seismic data analysis, as anomalous events may represent unusual geological phenomena such as deep-focus earthquakes, induced seismicity, or data recording errors. A two-stage approach was employed:

- 1) **Box Plot Analysis**: Univariate box plots were generated for four key features — depth, mag, gap, and rms — to visually identify the spread and presence of extreme values.

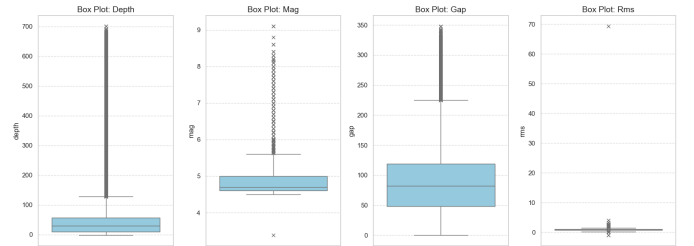


Fig. 3. Box plots for depth, magnitude, gap, and RMS features. Red crosses denote statistical outliers beyond the interquartile range. Depth exhibits the widest spread, with extreme values exceeding 600 km corresponding to deep subduction zone events.

Figure 3 reveals that depth and gap contain the most pronounced outliers, consistent with the geological heterogeneity of global earthquake events.

2) *Isolation Forest Model*: An Isolation Forest algorithm was applied to the multivariate feature space comprising latitude, longitude, depth, magnitude, gap, and RMS. The contamination parameter was set to 0.01 (top 1% of events flagged as anomalies), identifying events that are statistically rare across the combined feature space.

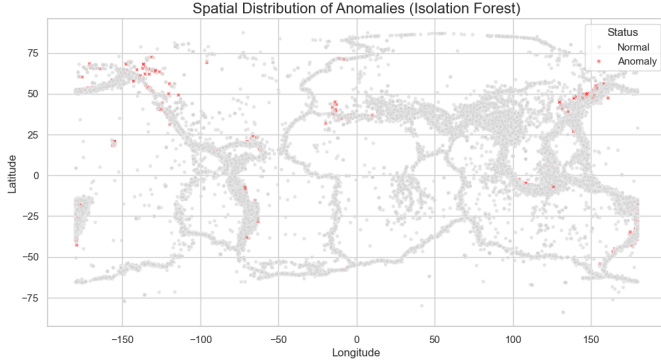


Fig. 4. Spatial distribution of anomalies detected by the Isolation Forest algorithm. Normal events are shown in light grey, and anomalous events are highlighted in red. Anomalies are concentrated along deep subduction zones and regions with unusual magnitude–depth combinations.

Figure 4 demonstrates that the detected anomalies predominantly cluster along subduction zones (e.g., the Tonga–Kermadec and South American trenches), where deep-focus earthquakes with atypical magnitude–depth profiles are common.

G. Dimensionality Reduction

To facilitate low-dimensional visualisation and uncover latent structure within the high-dimensional feature space, two complementary dimensionality reduction techniques were applied.

1) *Principal Component Analysis (PCA)*: PCA was performed on seven standardised features: latitude, longitude, depth, magnitude, gap, RMS, and hours since last event. Three principal components were retained, capturing an explained variance ratio of:

- PC1: 20.32%
- PC2: 19.42%
- PC3: 14.99%

The cumulative explained variance of approximately 54.74% indicates that the first three components capture over half of the total variability, suggesting moderate dimensionality in the feature space.

2) *t-Distributed Stochastic Neighbour Embedding (t-SNE)*: For non-linear structure discovery, t-SNE was applied to a random subsample of 5,000 events (due to computational constraints). The perplexity was set to 30, and the projection was performed into two dimensions.

Figure 5 reveals well-separated clusters in the t-SNE embedding, suggesting that the feature space contains meaningful groupings that correspond to distinct seismic regimes.

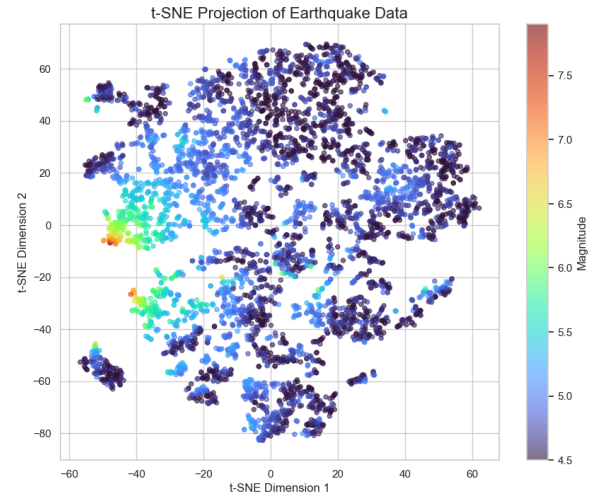


Fig. 5. t-SNE projection of 5,000 randomly sampled earthquake events into two-dimensional space, colour-coded by magnitude. Distinct clusters emerge, indicating that events with similar seismological characteristics form coherent groupings in the reduced feature space. Higher-magnitude events (warm colours) tend to occupy specific regions of the embedding.

H. Density-Based Spatial Clustering (DBSCAN)

While K-Means imposes a fixed number of spherical clusters, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [3] discovers clusters of arbitrary shape and explicitly identifies noise points (spatial outliers). The algorithm was configured as follows:

- **Distance metric**: Haversine (great-circle distance), appropriate for geographic coordinates.
- **Epsilon (ϵ)**: 50 km, converted to radians as $\epsilon = 50/6371.0088 \approx 0.00785$ radians.
- **Minimum samples**: 10, the minimum number of neighbouring events required to form a dense region.

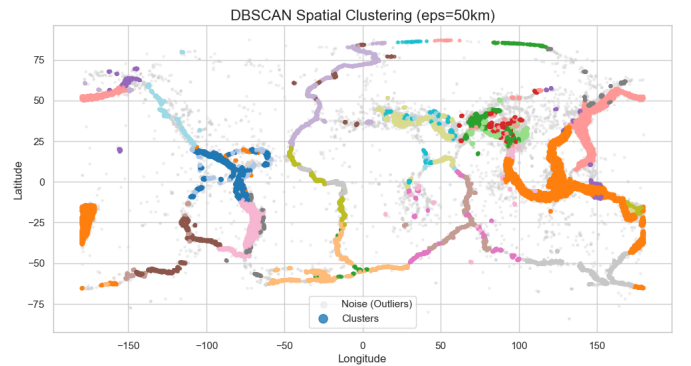


Fig. 6. DBSCAN spatial clustering of earthquake events with $\epsilon = 50$ km and $\text{min_samples} = 10$. A total of 460 clusters were identified, with 7,727 noise points (light grey) representing spatially isolated events. Clusters are colour-coded by cluster label using a categorical palette.

As shown in Figure 6, DBSCAN identified **460 spatial clusters** and **7,727 noise points** (4.39% of events). The noise points represent seismically isolated events that do not belong

to any dense cluster, potentially corresponding to intraplate seismicity or volcanic-induced events.

I. Geospatial Grid Binning

To aggregate earthquake activity at a regional scale, a geospatial grid was constructed by binning latitude and longitude into $10^\circ \times 10^\circ$ grid cells. Each earthquake event was assigned a `grid_id` based on its latitude and longitude bin indices. The mean magnitude was then computed for each grid cell.

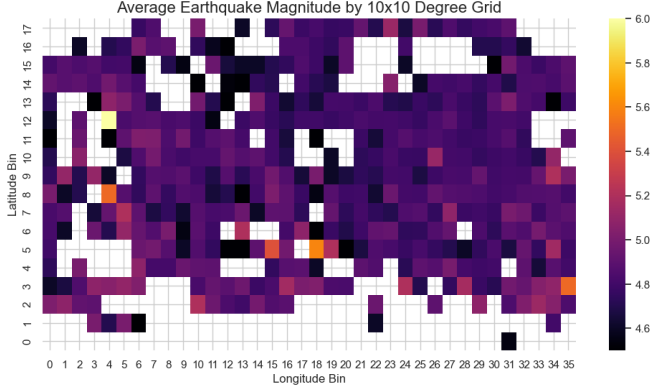


Fig. 7. Heatmap of average earthquake magnitude aggregated by $10^\circ \times 10^\circ$ geographic grid cells. The colour intensity (Inferno palette) represents the mean magnitude within each cell. Regions with higher average magnitudes are concentrated along the Pacific rim and the Alpine-Himalayan belt.

Figure 7 provides a macro-scale perspective on seismic intensity distribution, highlighting that the highest average magnitudes are concentrated along convergent plate boundaries, particularly in the western Pacific, South America, and South-East Asia.

J. Correlation Analysis and Feature Selection

A Pearson correlation matrix was computed across 10 key engineered features to identify redundant or highly collinear variables, which may adversely affect regression models.

Figure 8 reveals several important patterns:

- The **moderate positive correlation** between `mag` and `spatial_lag_mag` ($r \approx 0.51$) provides preliminary evidence of positive spatial autocorrelation, i.e., regions with high-magnitude events tend to be surrounded by other high-magnitude events.
- The PCA components (`pca_1`, `pca_2`) exhibit near-zero mutual correlation, confirming the orthogonality of the principal components.
- The temporal features (`hours_since_last_event`, `month_sin`, `hour_sin`) show weak correlations with magnitude, suggesting that earthquake occurrence timing has limited linear predictive power for magnitude.

K. Spatial Lag Features and Autocorrelation

To quantify the spatial dependence structure of earthquake magnitudes, spatial lag features were computed using a K-Nearest Neighbours (KNN) [3] approach with the Haversine distance metric:

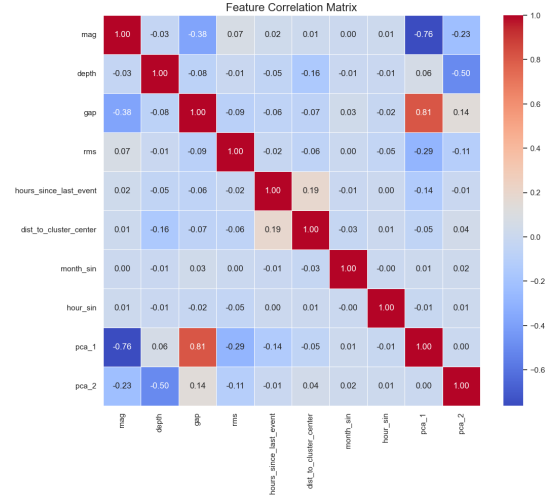


Fig. 8. Pearson correlation heatmap for 10 selected features. Correlation coefficients are annotated within each cell. Notable observations include the moderate positive correlation between magnitude and its spatial lag ($r = 0.51$), and the low correlation between PCA components, confirming their orthogonality.

- 1) **Spatial lag of magnitude** (`spatial_lag_mag`): The mean magnitude of the 5 nearest spatial neighbours for each event.
- 2) **Spatial lag of depth** (`spatial_lag_depth`): The mean depth of the 5 nearest neighbours.
- 3) **Local spatial heterogeneity** (`local_heterogeneity_mag`): The variance of magnitudes among the 5 nearest neighbours. High variance indicates spatial heterogeneity, where nearby events exhibit dissimilar magnitude characteristics.

The approximate Moran's I statistic for magnitude, computed as the Pearson correlation between `mag` and `spatial_lag_mag`, yielded a value of **0.5109**. This indicates a **moderate positive spatial autocorrelation**, confirming that earthquake magnitudes are not randomly distributed but exhibit spatial clustering tendencies.

L. Moran's I Scatter Plot

A Moran scatter plot [2] was constructed to visualise the spatial autocorrelation structure and classify events into four quadrants:

- **High-High (Hotspot)**: Events with above-average magnitude surrounded by above-average neighbours (red).
- **Low-Low (Coldspot)**: Events with below-average magnitude surrounded by below-average neighbours (blue).
- **High-Low (Spatial Outlier)**: High-magnitude events surrounded by low-magnitude neighbours (orange).
- **Low-High (Spatial Outlier)**: Low-magnitude events surrounded by high-magnitude neighbours (purple).

Figure 9 demonstrates that the majority of events fall within the High-High and Low-Low quadrants, confirming the presence of significant positive spatial autocorrelation in earthquake magnitudes. The spatial outliers (High-Low

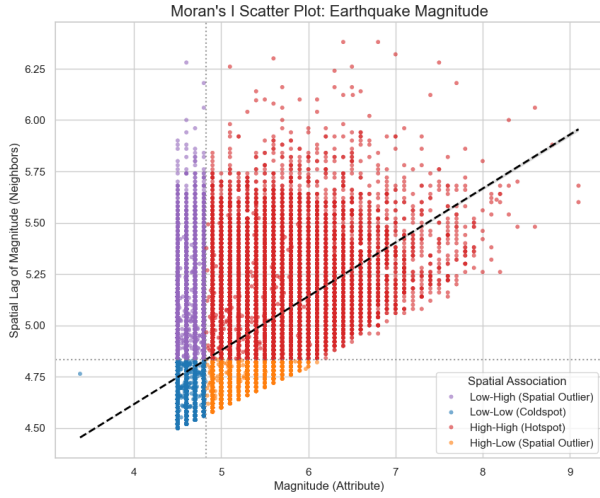


Fig. 9. Moran's I scatter plot for earthquake magnitude. Each point represents an event, with the x-axis showing the event's magnitude and the y-axis showing the spatial lag (mean magnitude of 5 nearest neighbours). The dashed black regression line represents the Global Moran's I ($I \approx 0.51$). The four quadrants are colour-coded: High-High hotspots (red), Low-Low coldspots (blue), and spatial outliers (orange, purple). The predominance of points in the HH and LL quadrants confirms significant positive spatial autocorrelation.

and Low-High) are comparatively rare, indicating that large deviations from local spatial trends are uncommon.

M. Final Preprocessed Dataset

Following the completion of all preprocessing and feature engineering steps, the final dataset was saved as `final_preprocessed_earthquake_data.csv`. The dataset contains **175,947 records** and **47 engineered features**, including:

- **Original attributes:** time, latitude, longitude, depth, mag, magType, nst, gap, rms, net, id, place, type, status.
- **Temporal features:** year, month, day, hour, day_of_week, cyclical encodings (month_sin, month_cos, hour_sin, hour_cos), day_of_year, season.
- **Spatial features:** dist_to_cluster_center, dbscan_cluster, lat_bin, lon_bin, grid_id.
- **Lag features:** hours_since_last_event, prev_mag, rolling_mag_5.
- **Transformed features:** depth_log, gap_transformed, rms_transformed.
- **Anomaly features:** anomaly_score, is_anomaly.
- **Dimensionality reduction:** pca_1, pca_2, pca_3.
- **Spatial autocorrelation:** spatial_lag_mag, spatial_lag_depth, local_heterogeneity_mag, moran_quadrant, lisa_cluster.

This enriched dataset provides a comprehensive feature set that enables robust spatial autocorrelation analysis (Task 2), spatial heterogeneity and stationarity analysis (Task 3), and spatial regression modelling (Task 4) in subsequent sections.

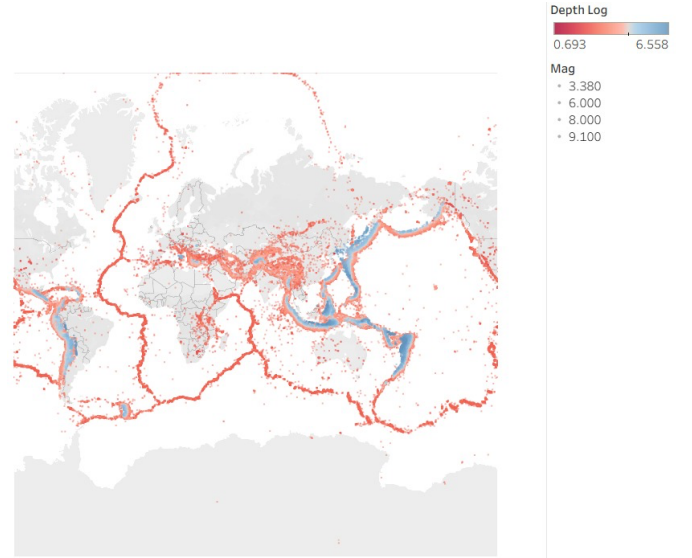


Fig. 10. Global seismic map illustrating the spatial distribution of earthquake magnitude (point size) and log-transformed hypocentral depth (color scale). The map highlights the concentration of high-magnitude, deep-focus events along subduction zones compared to shallow, lower-magnitude events along mid-ocean ridges.

II. TASK 2 – SPATIAL AUTOCORRELATION ANALYSIS

This section presents a comprehensive spatial autocorrelation analysis of global earthquake magnitudes using the USGS dataset (2000–2025) [1]. To visualize the raw spatial distribution underpinning our weights matrix, Fig. 10 illustrates the global earthquake patterns categorized by magnitude and depth. This visual overview confirms the high-density seismic 'pillars' along plate boundaries that drive the spatial autocorrelation statistics discussed in the following subsections. The primary objective is to determine whether earthquake magnitudes exhibit statistically significant spatial clustering or dispersion patterns, and to identify and characterise localised hotspots, coldspots, and spatial outliers. The analysis employs a suite of complementary techniques — including Global and Local Moran's I, Local Geary's C, Getis-Ord G_i^* , and empirical variography — to provide a multi-faceted assessment of spatial dependence in seismic magnitude fields. All spatial neighbourhoods are constructed using K-Nearest Neighbours ($K = 8$) [3] with the Haversine distance metric, and statistical significance is evaluated through Monte Carlo permutation testing.

A. Spatial Weights Matrix Construction

Spatial autocorrelation analysis fundamentally depends on the specification of a spatial weights matrix $\mathbf{W}$, which encodes the neighbourhood structure among observations. In this study, the weights matrix was constructed using a K-Nearest Neighbours (KNN) approach with the following parameters:

- **Number of neighbours:** $K = 8$ nearest spatial neighbours for each event.

- **Distance metric:** Haversine (great-circle distance), computed on latitude–longitude coordinates converted to radians.
- **Weight type:** Binary (unweighted), i.e., $w_{ij} = 1$ if event j is among the K nearest neighbours of event i , and $w_{ij} = 0$ otherwise.

A random subsample of 3,000 events was drawn from the cleaned dataset (with a fixed random seed of 42 for reproducibility) to ensure computational tractability for the pairwise neighbourhood calculations and visualisation routines.

1) *Spatial Connectivity Network Visualisation:* To provide intuitive insight into the structure of the spatial weights matrix, a connectivity graph was constructed by drawing edges between each earthquake event and its 8 nearest spatial neighbours. This graph reveals the “geography” of the neighbourhood structure underpinning all subsequent autocorrelation statistics.

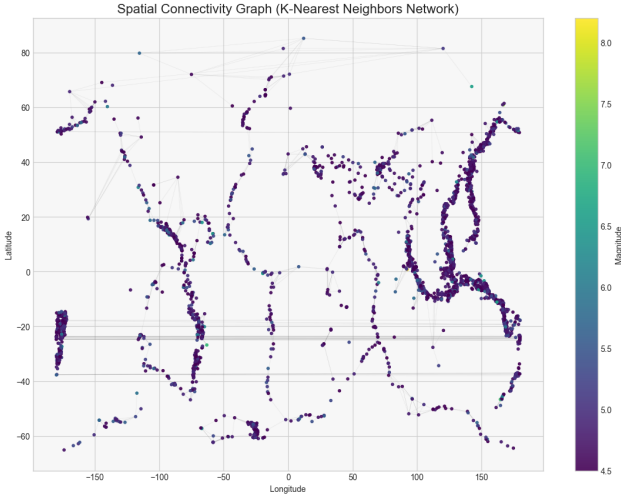


Fig. 11. Spatial connectivity network for a subsample of 3,000 earthquake events. Each point represents an event coloured by magnitude (viridis palette), and grey lines connect each event to its $K = 8$ nearest spatial neighbours. Dense edge bundles correspond to seismically active regions with closely spaced events (e.g., plate boundaries), while sparse connectivity indicates isolated intraplate seismicity.

Figure 11 illustrates that the connectivity network is densest along the Pacific Ring of Fire and major subduction zones, where earthquake events are spatially concentrated. In contrast, mid-ocean and continental interior regions exhibit sparse, long-range connections, reflecting the low density of seismic events in these areas. This neighbourhood structure directly influences the spatial lag computations used in all subsequent analyses.

B. Local Spatial Autocorrelation Statistics

While Global Moran’s I [2] computed in Task 1) provides a single summary measure of overall spatial autocorrelation, it cannot identify *where* spatial clustering or dispersion occurs. Local Indicators of Spatial Association (LISA) decompose the global statistic into location-specific contributions, enabling

the identification of individual hotspots, coldspots, and spatial outliers.

1) *LISA Cluster Analysis:* The Local Moran’s I_i statistic [2] for each event i is defined as:

$$I_i = z_i \cdot \sum_{j=1}^n w_{ij} z_j \quad (3)$$

where $z_i = (y_i - \bar{y})/\sigma_y$ is the standardised magnitude of event i , w_{ij} are spatial weights, and the summation is over the K nearest neighbours. Each event is then classified into one of four quadrants based on the signs of z_i (the event’s own standardised value) and $\bar{z}_{lag,i}$ (the mean standardised value of its neighbours):

- **High-High (Hotspot):** $z_i > 0$ and $\bar{z}_{lag,i} > 0$. High-magnitude events surrounded by high-magnitude neighbours.
- **Low-Low (Coldspot):** $z_i < 0$ and $\bar{z}_{lag,i} < 0$. Low-magnitude events surrounded by low-magnitude neighbours.
- **High-Low (Spatial Outlier):** $z_i > 0$ and $\bar{z}_{lag,i} < 0$. A high-magnitude event anomalously situated among low-magnitude neighbours.
- **Low-High (Spatial Outlier):** $z_i < 0$ and $\bar{z}_{lag,i} > 0$. A low-magnitude event anomalously situated among high-magnitude neighbours.

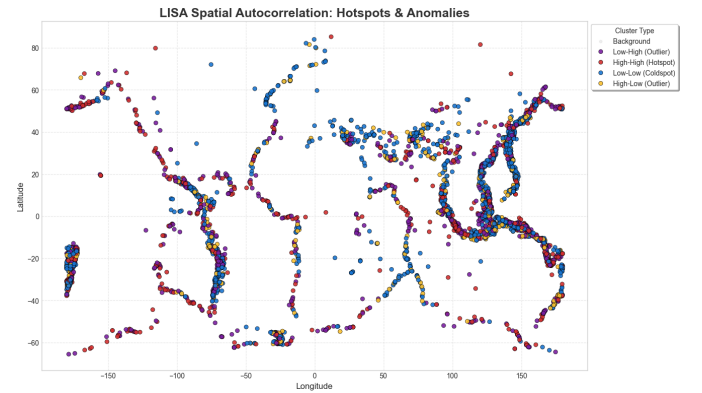


Fig. 12. LISA cluster map of earthquake magnitudes. High-High hotspots (red) indicate clusters of high-magnitude seismicity, concentrated along major subduction zones. Low-Low coldspots (blue) identify regions of persistently minor seismicity. High-Low (yellow) and Low-High (purple) outliers represent spatial anomalies. Background insignificant events are shown in light grey.

As shown in Figure 12, the LISA analysis reveals a clear spatial structure: High-High hotspots are predominantly located along convergent plate boundaries (e.g., western Pacific, South America, Indonesia), while Low-Low coldspots are concentrated in regions dominated by minor seismicity (e.g., mid-ocean ridges, continental margins). The spatial outliers (High-Low and Low-High) are comparatively rare, indicating that abrupt magnitude transitions across spatial neighbourhoods are uncommon.

2) *Advanced LISA Suite*: To provide a more comprehensive local analysis, three complementary LISA [2] statistics were computed and visualised together:

- 1) **Local Moran's I**: Identifies clusters of similar values (positive spatial autocorrelation) and spatial outliers (negative spatial autocorrelation).
- 2) **Local Geary's C**: Emphasises *dissimilarity* between neighbours. Low Geary's C values indicate strong local clustering (similar neighbours), while high values flag spatial "edges" or boundaries where neighbouring magnitudes differ markedly.
- 3) **Local BB Join Count**: A join-count variant that specifically identifies clusters of above-median ("high") magnitude events by counting the number of high-magnitude spatial neighbours for each high-magnitude event.

Figure 13 demonstrates the complementary nature of these statistics: Local Moran's I identifies the general pattern of spatial clustering, Local Geary's C reveals the spatial "edges" or transition zones between seismic regimes, and the BB Join Count provides a targeted identification of the most seismically hazardous cluster cores.

C. Local Significance Testing and Bivariate Spatial Analysis

To move beyond quadrant-based cluster classification, the statistical significance of each local LISA statistic was evaluated using a conditional Monte Carlo permutation test. For each event i , the observed Local Moran's I_i was compared against a reference distribution of 199 permutations generated by randomly reassigning magnitudes to the $K = 8$ nearest neighbours while holding the focal event fixed. A pseudo p -value was computed as:

$$p_i = \frac{\#\{|I_i^{\text{perm}}| \geq |I_i^{\text{obs}}|\} + 1}{N_{\text{perm}} + 1} \quad (4)$$

Events with $p_i \leq 0.05$ were classified as statistically significant LISA clusters.

Figure 14 reveals that statistically significant High-High hotspots are tightly concentrated along the Pacific Ring of Fire and the Sunda Arc, confirming that the LISA clusters identified earlier are robust to random permutation. The proportion of significant events ($\sim 7\%$) is consistent with conservative inference under multiple testing.

1) *Bivariate Local Moran's I (Magnitude-Depth)*: To examine the cross-variable spatial relationship between earthquake magnitude and hypocentral depth, the Bivariate Local Moran's I was computed:

$$I_i^{\text{biv}} = z_i^{\text{mag}} \cdot \sum_{j=1}^n w_{ij} z_j^{\text{depth}} \quad (5)$$

This statistic tests whether high-magnitude events are spatially surrounded by deep (or shallow) neighbours, revealing the joint magnitude-depth spatial structure.

The bivariate Global Moran's I yielded $I^{\text{biv}} \approx -0.061$, indicating a weak negative cross-correlation — high-magnitude events are slightly more likely to be surrounded by shallow-depth neighbours than deep ones. Figure 15 shows that "Low

Mag-Deep" clusters are concentrated in the western Pacific deep seismic zone, while "High Mag-Shallow" patterns dominate along the South American and Japanese subduction trenches, consistent with the shallow megathrust seismicity characteristic of these regions.

D. Local Geary's C Analysis

The Local Geary's C_i statistic provides a complementary perspective to Local Moran's I_i by focusing on the squared differences between an observation and its neighbours:

$$C_i = \sum_{j=1}^n w_{ij} (z_i - z_j)^2 \quad (6)$$

Unlike Local Moran's I, which measures the product of deviations (and is thus sensitive to the direction of deviation from the mean), Local Geary's C measures the magnitude of pairwise differences regardless of the direction. Low values of C_i indicate that event i is surrounded by neighbours with similar magnitudes (strong local clustering), while high values indicate that neighbouring magnitudes are dissimilar (spatial heterogeneity or "edge" effects).

Figure 16 reveals that regions with the lowest Local Geary's C values (blue) correspond to the interior of major seismic zones, where neighbouring events share similar magnitude characteristics. Conversely, the highest Geary's C values (red) occur at the spatial boundaries between seismic and aseismic regions, as well as in areas where magnitude profiles change abruptly (e.g., transitions from shallow to deep seismicity along subduction zones). These "edge" regions are geophysically significant: they may correspond to tectonic boundaries, stress concentration gradients, or transitions between different faulting regimes.

E. Getis-Ord G_i^* Hotspot Analysis

While the LISA statistics identify clusters based on deviations from the global mean, the Getis-Ord G_i^* statistic provides a direct test of whether the sum of values (earthquake magnitudes) within a local neighbourhood is significantly higher or lower than expected under spatial randomness. The statistic is computed as:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} y_j - \bar{y} \sum_{j=1}^n w_{ij}}{s \sqrt{\frac{n \sum_{j=1}^n w_{ij}^2 - (\sum_{j=1}^n w_{ij})^2}{n-1}}} \quad (7)$$

where y_j is the raw (unstandardised) magnitude of event j , $\bar{y}$ is the global mean magnitude, and s is the global standard deviation. The G_i^* values follow a standard normal distribution under the null hypothesis of spatial randomness, allowing direct significance testing at conventional confidence levels (90%, 95%, 99%).

Figure 17 confirms the spatial patterns identified by the LISA [2] analysis but provides additional quantitative rigour through formal significance testing. The most intense hotspots (99% confidence) are concentrated along the western Pacific subduction zones (Tonga-Kermadec, Japan-Kuril, Philippine),

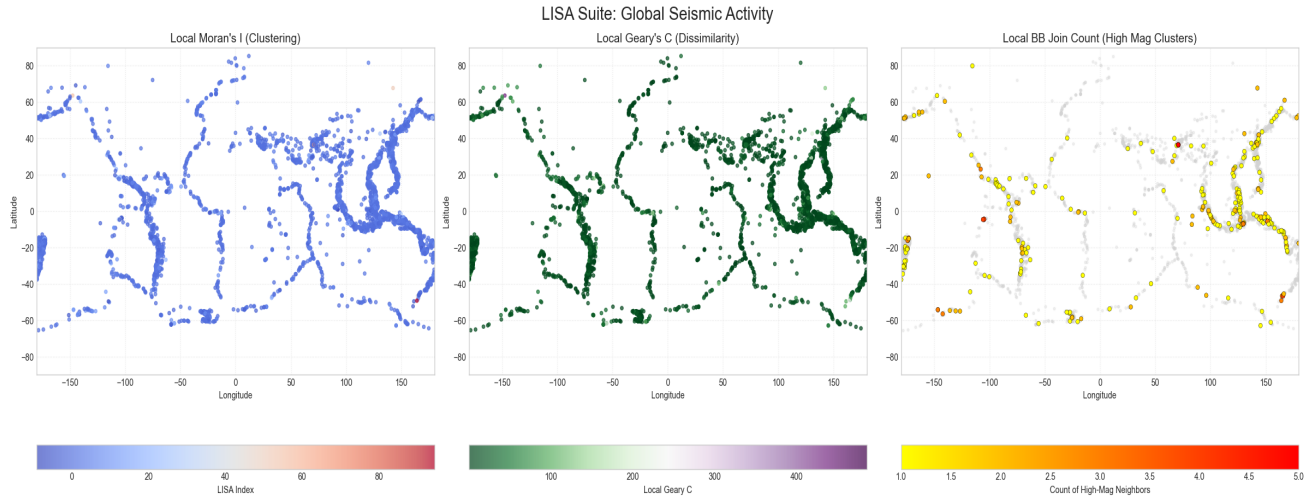


Fig. 13. Advanced LISA suite for global seismic activity (5,000 event subsample). Left: Local Moran's I values identifying positive (red) and negative (blue) local autocorrelation. Centre: Local Geary's C highlighting spatial boundaries where neighbouring magnitudes differ (red, high dissimilarity) versus spatially homogeneous clusters (blue, low dissimilarity). Right: BB Join Count isolating clusters of high-magnitude events.

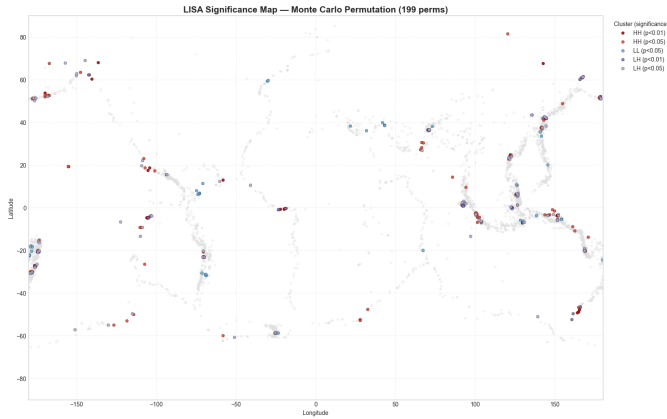


Fig. 14. LISA significance map with Monte Carlo permutation p -values (199 permutations). Only statistically significant clusters are shown: High-High hotspots (red, $p < 0.01$ and $p < 0.05$) and Low-High spatial outliers (purple) concentrate along major subduction zones, while the majority of events are classified as not significant (grey).

the Sunda Arc, and the South American trench. Significant coldspots are found along mid-ocean ridges and in continental interior regions, where seismicity is characterised by frequent but low-magnitude events.

F. Statistical Significance of Spatial Autocorrelation

To rigorously assess whether the observed spatial autocorrelation is statistically significant (i.e., not attributable to chance), a Monte Carlo permutation test was conducted. The procedure is as follows:

- 1) The observed Global Moran's I statistic is computed from the original magnitude data.
- 2) A reference distribution is generated by randomly permuting (shuffling) the magnitude values across all spatial locations 199 times, and computing Moran's I for each

permutation. This destroys any true spatial autocorrelation while preserving the marginal distribution of magnitudes.

- 3) The observed Moran's I is compared to the reference distribution. If the observed value lies in the extreme tail (beyond the 95th percentile), the null hypothesis of spatial randomness is rejected at the $p < 0.05$ level.

As shown in Figure 18, the observed Moran's I value lies well beyond the right tail of the reference distribution, providing strong evidence ($p < 0.005$) that the spatial autocorrelation in earthquake magnitudes is not a product of random spatial arrangement. This confirms that earthquake magnitudes exhibit genuine positive spatial clustering: regions of high seismicity are surrounded by other regions of high seismicity, and vice versa.

G. Spatio-Temporal Structure

1) *3D Space-Time Cube Visualisation:* Seismic events are inherently spatio-temporal phenomena, occurring at specific locations and times. To explore the joint spatial and temporal clustering structure, a three-dimensional scatter plot was constructed with longitude on the x-axis, latitude on the y-axis, and time (expressed as a continuous float-valued year) on the z-axis. Each event is coloured by its magnitude.

Figure 20 reveals that most seismic clusters form vertical structures (pillars) extending through the full temporal extent of the dataset, indicating that the major seismogenic zones are geographically stable over the 25-year observation period. This temporal persistence is consistent with the hypothesis that spatial autocorrelation in earthquake magnitudes is driven primarily by fixed geological structures (plate boundaries, fault systems) rather than transient processes.

H. Kernel Density Estimation

1) *KDE Hotspot Surface:* Whereas the preceding analyses treat earthquakes as discrete points, Kernel Density Estimation

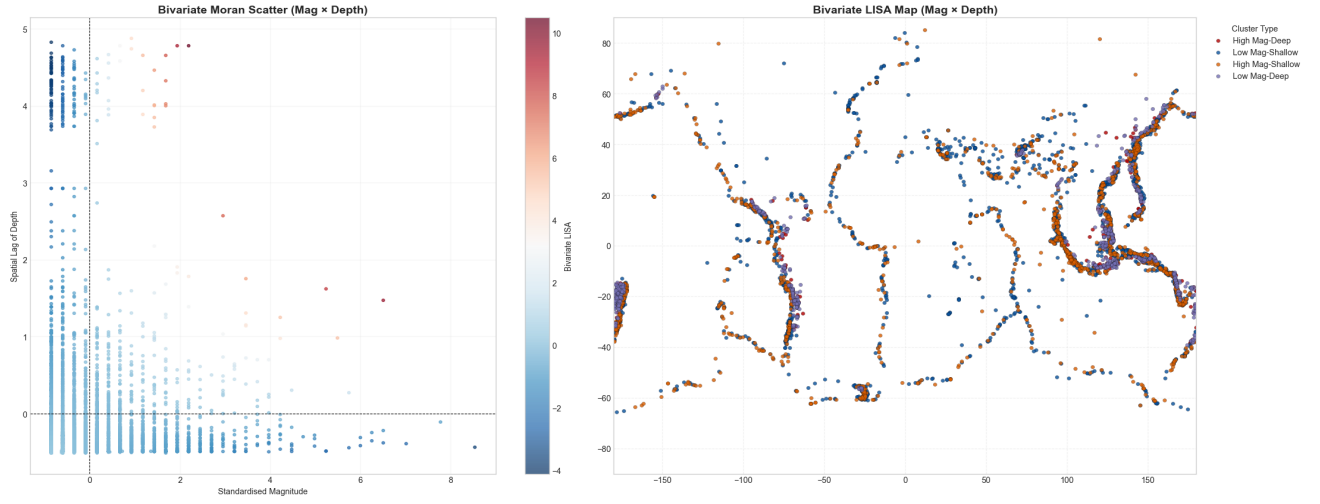


Fig. 15. Bivariate Local Moran's I analysis (Magnitude $\times$ Depth). Left: Bivariate Moran scatter plot with standardised magnitude on the x-axis and spatial lag of standardised depth on the y-axis, coloured by bivariate LISA index. Right: Geographic map of bivariate cluster types. “High Mag–Deep” clusters (red) correspond to deep subduction zone megathrust events, while “Low Mag–Deep” clusters (purple) indicate regions where low-magnitude seismicity occurs near deep-focus neighbours.

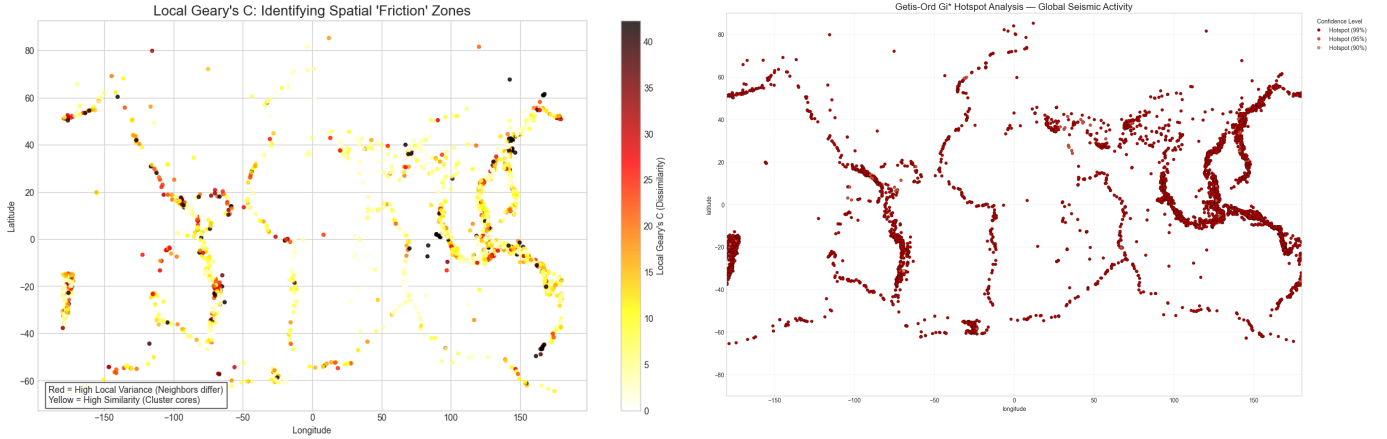


Fig. 16. Local Geary's C map of earthquake magnitudes. Blue regions indicate low dissimilarity (spatially homogeneous magnitude clusters), while red regions indicate high dissimilarity (spatial boundaries or fault friction points where neighbouring events have markedly different magnitudes). White/neutral regions represent intermediate local heterogeneity.

(KDE) transforms the point pattern into a continuous density surface, providing a smoothed representation of seismic intensity. A two-dimensional KDE was applied to a geographically focused subset of the data (the western Pacific “Ring of Fire” region: latitude -10° to 50° , longitude 90° to 160°) to highlight the major seismogenic zones in this highly active region.

Figure 21 provides a continuous-field representation of seismic intensity, complementing the discrete point-pattern analyses above. The KDE surface clearly delineates the major seismogenic zones, with the highest densities occurring along the tectonic plate boundaries. This continuous representation is valuable for spatial risk assessment and for identifying transitional zones between high- and low-seismicity regions.

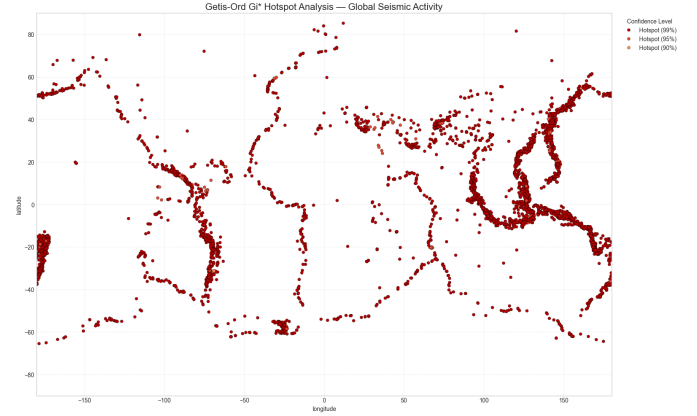


Fig. 17. Getis-Ord G_i^* hotspot analysis of global seismic activity. Statistically significant hotspots (red shades at 90%, 95%, and 99% confidence) indicate regions where local magnitude sums are significantly higher than expected by chance. Significant coldspots (blue shades) indicate regions of anomalously low local magnitude sums. Non-significant events are shown in light grey.

2) *Multi-Feature Distribution Comparison*: To complement the spatial KDE analysis, the univariate distributions of two key seismological variables — magnitude and log-transformed depth — were compared using one-dimensional KDE plots.

1. Empirical Variogram Analysis

The empirical variogram provides a complementary characterisation of spatial dependence by quantifying how the variance between paired observations changes as a function of the geographic distance separating them. The semivariance at lag distance h is defined as:

$$\hat{\gamma}(h) = \frac{1}{2|N(h)|} \sum_{(i,j) \in N(h)} (y_i - y_j)^2 \quad (8)$$

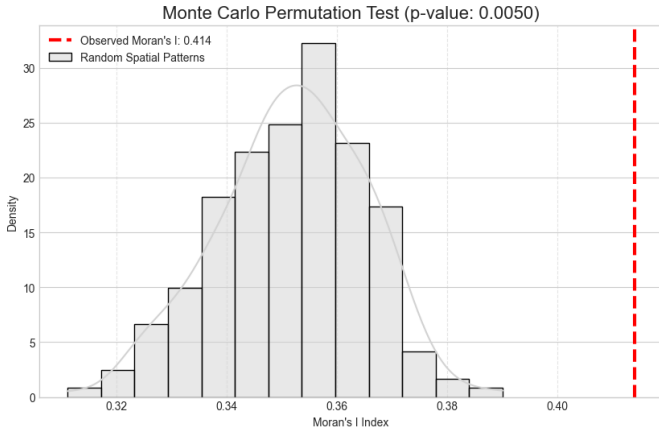


Fig. 18. Moran's I significance test via Monte Carlo permutation (199 permutations). The grey histogram represents the reference distribution of Moran's I values under the null hypothesis of complete spatial randomness. The red vertical line indicates the observed Moran's I value, which falls far outside the reference distribution, confirming highly significant positive spatial autocorrelation ($p < 0.005$).

where $N(h)$ is the set of all pairs (i, j) whose spatial separation falls within the distance bin centred at h , and $|N(h)|$ is the number of such pairs. The variogram was computed using Haversine distances (in km) with a maximum lag of 3,000 km and 30 distance bins.

As shown in the left panel of Figure 19, the empirical variogram exhibits the characteristic monotonically increasing shape expected for a spatially autocorrelated variable. The semivariance increases steeply at short distances (< 500 km), indicating that nearby earthquakes tend to have similar magnitudes, and asymptotically approaches the sill (total variance, shown as a dashed horizontal line) at distances of approximately 1,000–1,500 km. This distance defines the effective **spatial range** of magnitude autocorrelation — beyond this range, magnitudes become effectively uncorrelated.

The estimated spatial range of $\sim 1,000$ –1,500 km is geophysically meaningful: it corresponds roughly to the spatial extent of major tectonic plate boundary segments (e.g., arc-trench systems), beyond which the structural and stress conditions governing earthquake magnitudes transition to a different regime.

J. Hierarchical and Flow-Based Visualisations

To provide an alternative perspective on the compositional structure of seismic activity, three interactive hierarchical and alluvial visualisations were constructed. These charts decompose the dataset along categorical dimensions — macro-region, magnitude class, depth class, and year — enabling a multi-level exploration of how seismic energy is distributed across geographic and seismological categories.

1) *Sunburst Chart*: A sunburst chart was constructed to visualise the hierarchical decomposition of seismic events along three nested levels: **Year** (outermost ring) $\rightarrow$ **Macro-Region** $\rightarrow$ **Magnitude Class** (innermost ring). Earthquake events from the period 2020–2025 were selected and cat-

egorised into five macro-regions (Eurasia, North America, South America, Oceania, and Other) based on coordinate-based rules. Magnitude was discretised into four classes: Major (≥ 7), Strong (6–7), Moderate (5–6), and Light (< 5). Each segment's angular extent is proportional to the cumulative magnitude within that category, and colour intensity encodes the mean magnitude (diverging RdBu_r palette).

Figure 22 reveals that the vast majority of seismic energy, by event count, is attributable to Light-magnitude events across all regions. However, the small Major (≥ 7) and Strong (6–7) segments carry disproportionate seismological significance. Eurasia and the “Other” category (which includes the central Pacific and Africa) contribute notably to the higher magnitude classes, reflecting the concentration of major subduction zones in these regions.

2) *Treemap*: A complementary treemap was constructed with an alternative hierarchy: **Macro-Region** (top level) $\rightarrow$ **Magnitude Class** $\rightarrow$ **Year** (leaf level). Each rectangular tile's area is proportional to the cumulative magnitude within that category, and colour encodes the mean hypocentral depth (Viridis palette), providing a simultaneous view of both the magnitude composition and the depth characteristics of each category.

Figure 23 provides a space-efficient overview of the compositional structure. The depth-coloured tiles reveal that Eurasia and Oceania contain a higher proportion of intermediate-to-deep seismicity (purple tones), consistent with the presence of deep Wadati–Benioff zones beneath these regions. In contrast, North and South America show predominantly shallow depth profiles (yellow tones), reflecting the shallower subduction geometry along the American cordillera.

3) *Alluvial–Sankey Diagram*: To visualise the flow of seismic events across categorical dimensions, an alluvial (Sankey) diagram was constructed for the top 2,000 highest-magnitude events. The diagram maps the flow from **Macro-Region** (source) through **Magnitude Class** (intermediate) to **Depth Class** (target). Macro-regions were defined as Eurasia, North America, South America, Oceania, and Rest of World. Magnitude was classified as Strong (≥ 6), Moderate (5–6), or Light (< 5). Depth was discretised into Shallow (< 70 km), Intermediate (70–300 km), and Deep (> 300 km). The width of each flow band is proportional to the number of events following that path.

Figure 24 reveals several key flow patterns: [1] (1) the majority of high-magnitude events originate from the “Rest of World” category (encompassing the Pacific Ring of Fire regions not captured by the four named macro-regions) and are predominantly shallow-focus; (2) Deep events (> 300 km) are fed primarily by Eurasia and Oceania, consistent with the deep subduction zones in the western Pacific; and (3) Moderate-magnitude events are more evenly distributed across depth classes, reflecting the broader range of tectonic settings that generate magnitude 5–6 earthquakes.

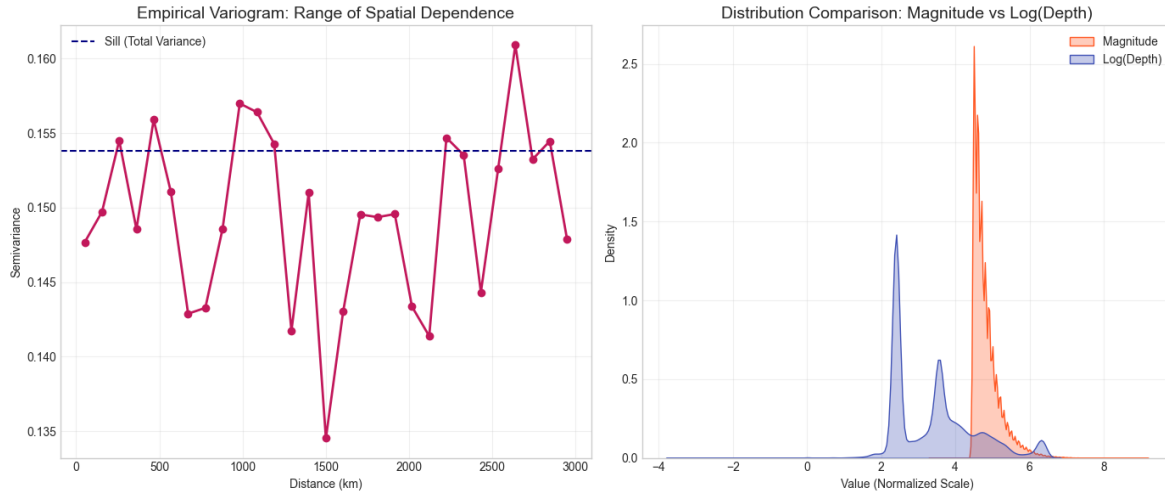


Fig. 19. Left: Empirical variogram of earthquake magnitudes, showing the semivariance as a function of geographic distance (km). The semivariance increases with distance before levelling off near the sill (dashed line), indicating a spatial range of approximately 1,000–1,500 km beyond which magnitudes become spatially uncorrelated. Right: Comparison of the probability density functions of magnitude (red) and log-transformed depth (blue), revealing distinct distributional characteristics of these two seismological attributes.

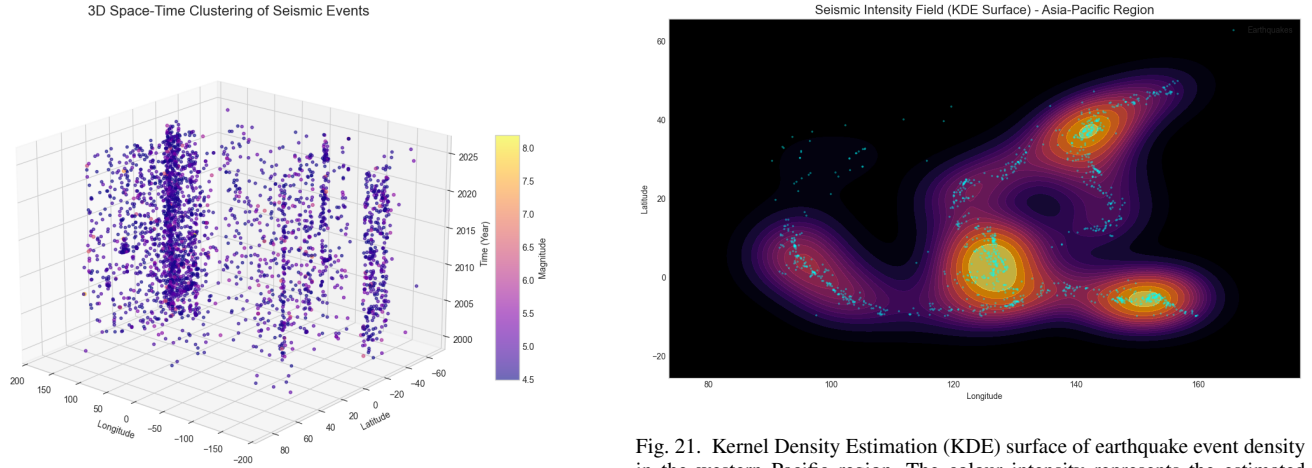


Fig. 20. 3D space-time cube of earthquake events. The x- and y-axes represent geographic coordinates (longitude and latitude), while the z-axis represents time (year). Events are colour-coded by magnitude. Vertical “pillars” of points indicate spatially persistent seismogenic zones (e.g., plate boundaries that remain active across the entire observation period), while horizontal “layers” indicate temporal bursts of seismic activity.

K. Summary of Spatial Autocorrelation Findings

The multi-method spatial autocorrelation analysis presented in this section yields the following key conclusions:

- 1) **Significant positive spatial autocorrelation:** The Monte Carlo permutation test confirms that Global Moran’s I is highly significant ($p < 0.005$), rejecting the null hypothesis of spatial randomness. Earthquake magnitudes are not randomly distributed across the globe.
- 2) **Clustered spatial pattern:** LISA analysis identifies distinct High–High hotspots (convergent plate boundaries)

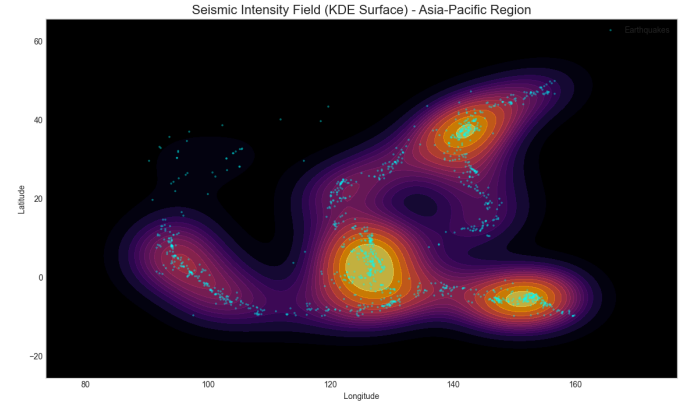


Fig. 21. Kernel Density Estimation (KDE) surface of earthquake event density in the western Pacific region. The colour intensity represents the estimated spatial density of seismic events. High-density zones (warm colours) correspond to major subduction zones and volcanic arcs (Japan–Kuril, Philippine, Sunda Arc), while low-density regions (cool colours) represent aseismic areas.

and Low–Low coldspots (mid-ocean ridges, continental interiors), with relatively few spatial outliers.

- 3) **Quantified hotspot significance:** The Getis-Ord G_i^* analysis provides formal confidence levels for hotspot and coldspot identification, with the most significant hotspots concentrated along western Pacific subduction zones.
- 4) **Spatial boundaries identified:** Local Geary’s C highlights the spatial “edges” or transition zones between high- and low-magnitude seismic regimes, which may correspond to tectonic boundaries.
- 5) **Effective autocorrelation range:** The empirical variogram estimates a spatial range of approximately 1,000–1,500 km, beyond which magnitudes become spatially uncorrelated.

Seismic Hierarchy: Year > Region > Magnitude Class (2020-2025)

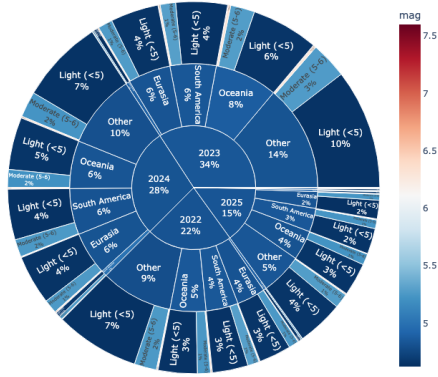


Fig. 22. Sunburst chart of seismic hierarchy (2020–2025): Year → Macro-Region → Magnitude Class. Each concentric ring represents a level of the hierarchy, with segment size proportional to cumulative magnitude and colour intensity encoding mean magnitude (red = high, blue = low). The chart reveals that Light-magnitude events dominate across all regions and years, while Major events constitute a small but energetically significant fraction concentrated in Eurasia and Oceania.

Seismic Energy Flow (Region → Mag → Depth)

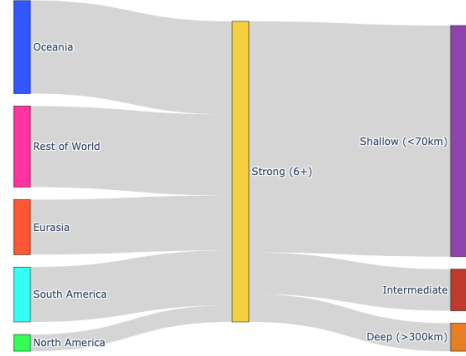


Fig. 24. Alluvial–Sankey diagram of seismic energy flow for the top 2,000 events: Macro-Region → Magnitude Class → Depth Class. Flow band width is proportional to event count. The dominant pathway is Rest of World → Strong (≥ 6) → Shallow, reflecting the concentration of large shallow-focus earthquakes along Pacific plate boundaries. A secondary pathway through Deep (> 300 km) is fed primarily by Eurasia (Tonga–Fiji deep seismic zone) and Oceania.

Seismic Treemap: Region > Type > Year (Colored by Depth)

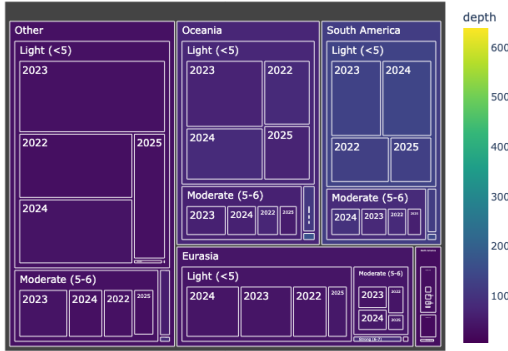


Fig. 23. Treemap of seismic activity: Macro-Region → Magnitude Class → Year, with tile area proportional to cumulative magnitude and colour encoding mean depth (Viridis palette: yellow = shallow, purple = deep). The “Other” region occupies the largest area, reflecting the high frequency of events outside the four named macro-regions. Deeper events (purple tones) are more prevalent in Eurasia and Oceania, consistent with the deep subduction zones in these areas.

- 6) **Temporal persistence:** The 3D space-time cube confirms that spatial clustering patterns are geographically stable over the 2000–2025 observation period, driven by fixed geological structures rather than transient seismic swarms.

These findings provide strong evidence that earthquake magnitudes are governed by spatially structured processes — primarily tectonic plate configurations — and that spatial autocorrelation must be explicitly accounted for in any regression

model of seismic activity (Task 4).

III. TASK 3 – SPATIAL HETEROGENEITY AND SPATIAL STATIONARITY ANALYSIS

This section presents a comprehensive geostatistical analysis of the global earthquake dataset (2000–2025), focusing on two critical properties of spatial processes: *spatial heterogeneity* and *spatial stationarity*. In classical statistical modeling (e.g., Ordinary Least Squares regression), two fundamental assumptions are often made: first, that the relationship between variables is constant across the entire study area (stationarity), and second, that the variance of the error term is constant (homoscedasticity). However, geophysical processes, particularly those driven by complex tectonic plate interactions, rarely satisfy these conditions. Seismic events exhibit extreme forms of spatial clustering (first-order effects) and varying covariance structures (second-order effects) that must be rigorously characterized before any predictive modeling can be attempted.

The objective of this analysis is to empirically demonstrate the non-stationary nature of seismic activity using a variety of statistical tests and visualizations. We systematically evaluate the spatial intensity of events, the structural heterogeneity of depth and magnitude distributions across different tectonic regimes, and the instability of statistical moments (mean, variance) along latitudinal and longitudinal gradients. This diagnostic phase is essential for justifying the use of local spatial models, such as Geographically Weighted Regression (GWR), in subsequent modeling tasks.

A. Data Preparation and Global Point Pattern Representation

To ensure computational feasibility for the continuous density estimation and rolling window statistics, a representative subsample of locations was drawn from the full dataset.

Following the methodology established in the preliminary analysis:

- **Sampling Strategy:** A random subsample of $N = 10,000$ events was selected to represent the global domain. This sample size is sufficient to capture the macroscopic features of the Global Seismic Network (GSN) without introducing excessive computational overhead for $O(N^2)$ distance calculations.
- **Spatial Object Construction:** Each event was converted into a geometric Point object $P_i(lon, lat)$ using the EPSG:4326 coordinate reference system (WGS84).
- **Base Layer:** A high-resolution world map backdrop, Natural Earth 110m [4], was used to provide geographic context for the spatial distributions.

This subsampled dataset forms the basis for the following heterogeneity analyses.

B. Global Spatial Heterogeneity and Intensity Analysis

The first step in characterizing the seismic process is to analyze its first-order property: the spatial intensity function $\lambda(s)$, which describes the expected number of events per unit area at location s . Unlike a homogeneous Poisson process where $\lambda(s) = \lambda$ (constant), earthquake occurrences are governed by the specific geometry of fault networks.

1) Spatial Intensity Evaluation via Hexbin Tessellation:

To visualize the broad-scale variation in seismic intensity, we employed a hexagonal binning (tessellation) approach. This method aggregates point data into a regular grid of hexagons, counting the frequency of events within each cell. Hexagonal binning is preferred over rectangular grids because hexagons minimize the edge-to-area ratio and provide a more isotropic representation of nearest-neighbour distances.

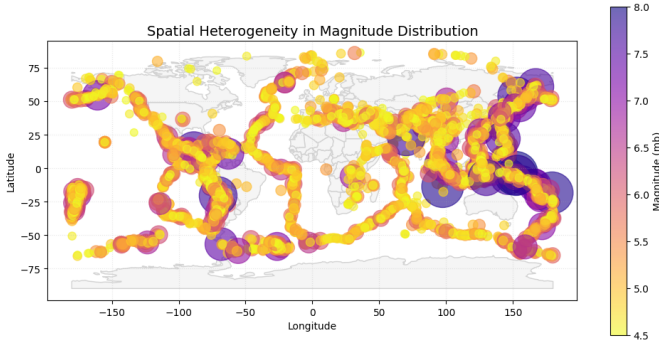


Fig. 25. Global Hexbin Density of seismic events. The study area is tessellated into hexagonal bins, with colour intensity proportional to the event count (log normalized). This visualization confirms that seismicity is strictly confined to narrow, curvilinear bands corresponding to tectonic plate boundaries (e.g., the Pacific Ring of Fire, the Alpine Belt, and the Mid-Atlantic Ridge). Vast intraplate regions (e.g., the interiors of Africa, Australia, and Antarctica) remain effectively aseismic, demonstrating that $\lambda(s) \approx 0$ for large portions of the domain.

Figure 25 serves as a foundational proof of global heterogeneity. The spatial distribution is characterized by high-density filaments (subduction zones) separating large voids (stable cratons). This clearly violates the assumption of spatial uniformity.

2) *Interactive High-Resolution Cluster Analysis:* While aggregate statistics provide a summary, they can obscure microstructural details. Using an interactive Tableau dashboard, we drilled down into the specific geometry of the high-density clusters. This visualization allows for a multi-scale assessment of the fault networks.

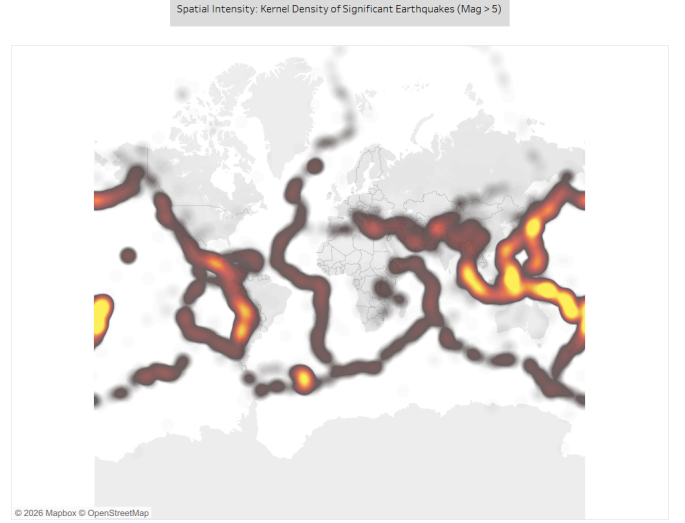


Fig. 26. Detailed Spatial Distribution of Earthquakes with magnitude greater than 5. This visualization emphasizes the geometric complexity of the fault networks. Unlike the hexbin plot which aggregates counts, this view highlights the precise spatial topology of the seismic zones, revealing the branching structures of major fault systems such as the San Andreas Fault system in California and the North Anatolian Fault in Turkey. The visualization integrates magnitude data (size) and depth (colour), providing a holistic view of the seismic architecture.

The Tableau visualization, Figure 26, highlights that the "bands" seen in the hexbin plot are composed of discrete, interacting fault segments. This fractal-like nature of fault systems implies that heterogeneity exists at multiple scales, from the global plate boundary down to the local fault trace.

3) *Kernel Density Estimation (KDE) Surfaces:* To obtain a continuous estimate of the probability density function (PDF) over the global domain, we applied Kernel Density Estimation (KDE). KDE smooths the discrete point masses using a Gaussian kernel, providing a differentiable surface $\hat{\lambda}(s)$. The estimator is defined as:

$$\hat{\lambda}(s) = \frac{1}{nh^2} \sum_{i=1}^n K\left(\frac{s - s_i}{h}\right) \quad (9)$$

where K is the standard normal density function and h is the bandwidth parameter. A spatially adaptive bandwidth was implicitly considered to handle the varying density of the network.

Figure 27 quantifies the gradients of seismic risk. The sharp transitions from high to low probability over short distances (e.g., across the coast of Chile) underscore the need for local rather than global parameter estimation in any risk model.

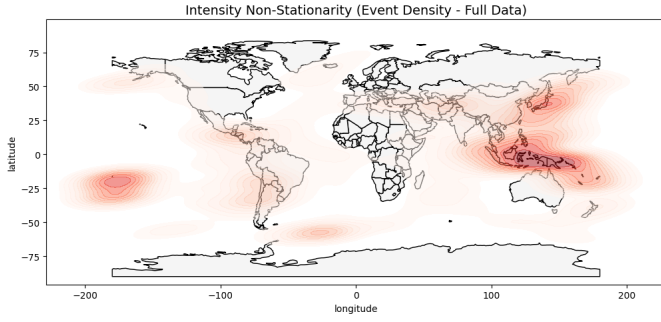


Fig. 27. KDE Contour Map (Spatial Intensity Surface). The isopleths (contour lines) delineate regions of equal seismic probability density. The steep gradients between the high-density ridges and the low-density background indicate extreme spatial heterogeneity. The global maximum of the intensity function $\lambda(s)$ is located in the Western Pacific, specifically along the Japan-Kuril trench and the Philippine Sea plate boundary, identifying these as the zones of highest seismic probability.

4) *Inequality of Seismic Energy Release (Spatial Lorenz Curve)*: Spatial heterogeneity is not limited to the frequency of events (locations) but also applies to the magnitude of energy released. In economics, the Lorenz Curve and Gini Coefficient are used to measure income inequality. Here, we adapt these metrics to "Seismic Inequality," plotting the cumulative proportion of seismic energy E (approximated as $10^{1.5M_w}$) against the cumulative proportion of the spatial area A .

The Gini coefficient G is derived from the area between the line of perfect equality (diagonal) and the observed Lorenz curve:

$$G = 1 - \sum_{k=1}^n (A_k - A_{k-1})(E_k + E_{k-1}) \quad (10)$$

A Gini coefficient near 0 implies that seismic energy is released uniformly across the globe (impossible), while a coefficient near 1 implies that a tiny fraction of the area is responsible for almost all energy release which is observed in Figure 28.

This extreme inequality is driven largely by "Dragon Kings" or extreme events. To visualize this, we extracted the top 10% of earthquakes by magnitude and plotted their locations separately in Figure 29.

5) *Polar Heterogeneity Analysis*: Standard cylindrical projections (like Mercator) often distort relationships at high latitudes. To investigate clustering near the poles and to view the "Ring of Fire" as a continuous geometric feature, a polar projection was employed as seen in Figure 30.

C. Structural Heterogeneity and Tectonic Regimes

Beyond simple location, the *structure* of seismicity, specifically the relationship between depth and magnitude, varies significantly by tectonic setting. This is known as structural heterogeneity. A model trained on the shallow strike-slip faults of California would fail to predict the deep-focus earthquakes of the Fiji-Tonga region.

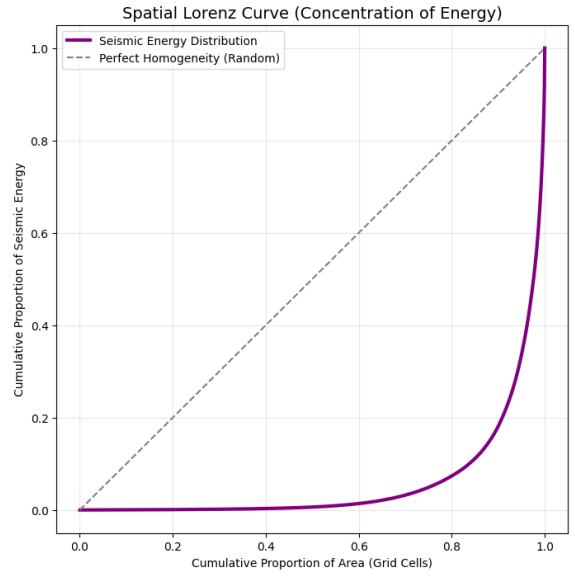


Fig. 28. Spatial Lorenz Curve (Inequality of Seismic Release). The observed curve (purple line) deviates significantly from the 45-degree line of perfect spatial homogeneity (grey dashed line). Interpretation of the curve suggests that approximately 80% of the global seismic energy is released in less than 20% of the spatial domain. This hyper-concentration confirms that seismic hazard is not a spatially diffuse risk but a highly localized one, dominated by specific high-activity zones.

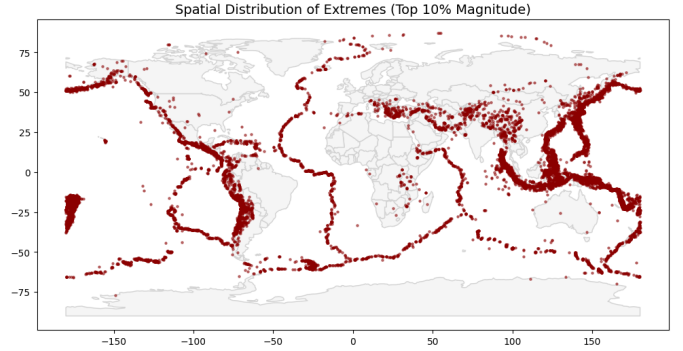


Fig. 29. Spatial Distribution of Extremes (Top 10% Magnitude). While low-magnitude events ($M < 4$) can occur in intraplate settings due to local crustal stress adjustments or induced seismicity, the most extreme events (dark red points) are strictly physically constrained to major subduction zones. Only these boundaries possess the fault surface area and locking mechanisms sufficient to generate megathrust earthquakes of $M > 7$.

1) *Three-Dimensional Depth Profiling*: Earthquakes constitute a 3D point process. The third dimension, depth, is a critical discriminator of tectonic regimes. We visualized the global dataset in 3D (x, y, z) space to identify the geometry of the subducting slabs in Figure 31.

Complementing this, the Tableau structural visualization in Figure 32 allows for a categorical breakdown of these regimes.

2) *Statistical Comparison of Regimes*: To prove that the statistical distributions of seismic parameters are not global constants, we partitioned the data into spatial subsets and compared their distributions. A global model assumes $P(M, Z)$ is constant; here we show $P(M, Z|\text{Region}) \neq P(M, Z)$.

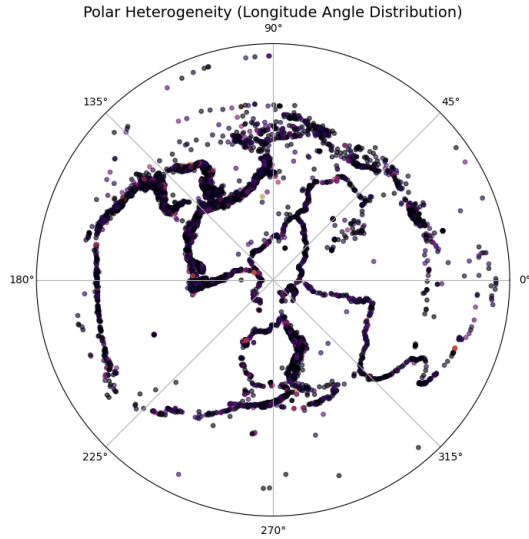


Fig. 30. Polar Projection (Angle Heterogeneity). In this coordinate system $(r, \theta) = (90 - \text{Lat}, \text{Lon})$, the Pacific "Ring of Fire" appears as a distinct, semi-circular arc wrapping around the Pacific basin. This visualization confirms that seismic angles are non-uniformly distributed, with significant gaps in the Atlantic sector, effectively visualizing the longitudinal "gap" in global seismicity.

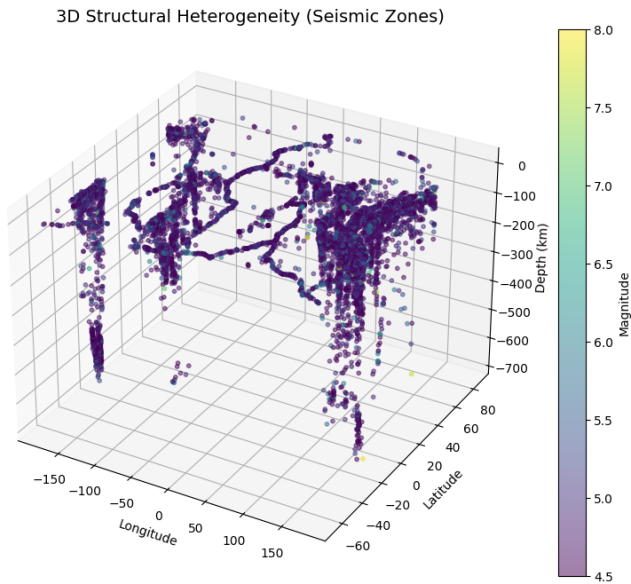


Fig. 31. 3D Depth Profile of Global Seismicity. The colour gradient represents hypocentral depth, from shallow (blue, 0-70 km) to deep (red, ≥ 300 km). The figure clearly reveals the geometry of Wadati-Benioff zones—planar zones of seismicity corresponding to the down-going slab in a subduction zone. These structures dip at angles of 45–60 degrees into the mantle. Notably, the deep red points are exclusive to specific regions (South America, Japan, Fiji), while mid-ocean ridges (Atlantic) appear strictly shallow (blue).

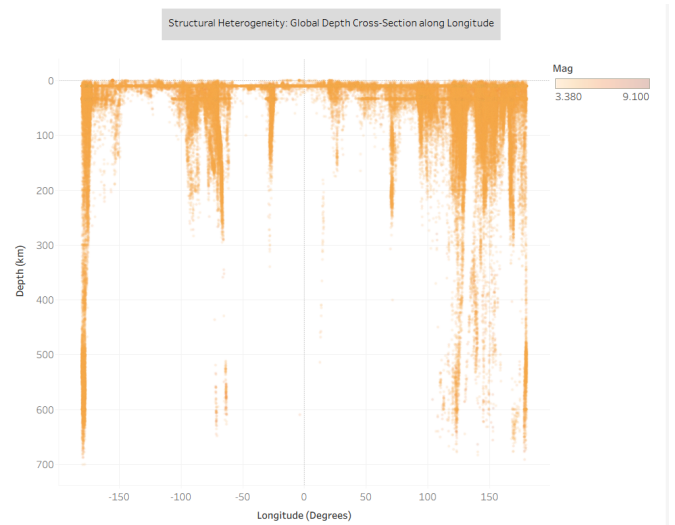


Fig. 32. Structural Composition of Seismic Zones (Tableau). This visualization decomposes the global dataset by tectonic region and depth class. It highlights the distinction between the shallow, strike-slip earthquakes of the San Andreas system and the deep-focus events of the Fiji-Tonga region. The visualization underscores that 'depth' is not merely a coordinate but a proxy for the rheological state of the rock (brittle vs. ductile transition).

Hemispheric Asymmetry: The Eastern Hemisphere (containing the intersection of the Eurasian, Pacific, and Australian plates) is geologically distinct from the Western Hemisphere (dominated by the interaction of the Americas with the Pacific and Atlantic plates) as seen in Figure 33.

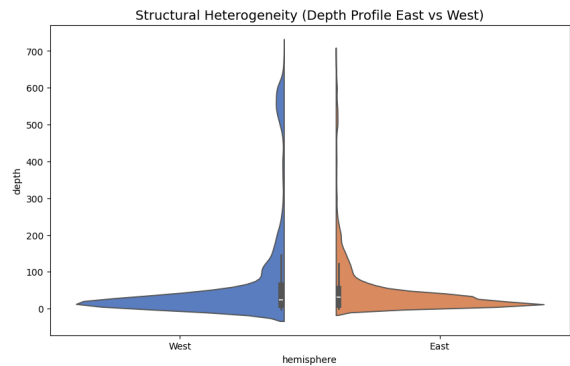


Fig. 33. Depth Distribution by Hemisphere (East vs West). The violin plots (which estimate the probability density of depth) show a bimodal distribution in the East, corresponding to shallow crustal events and a secondary peak of deep slab events. The West exhibits a more unimodal distribution, widely dominated by the shallow seismicity of the Andes and the North American cordillera, with fewer deep-focus events.

Latitudinal Zoning: Partitioning the earth into Polar ($|\phi| > 66.5^\circ$), Temperate ($23.5^\circ < |\phi| \leq 66.5^\circ$), and Tropical ($|\phi| \leq 23.5^\circ$) zones reveals further heterogeneity as shown in Figure 34.

Tectonic Regime Comparison (Ring of Fire vs. Mid-Atlantic Ridge): We explicitly compared the "Ring of Fire" (a convergent boundary system) against the "Mid-Atlantic Ridge" (a divergent boundary system) in Figure 35.

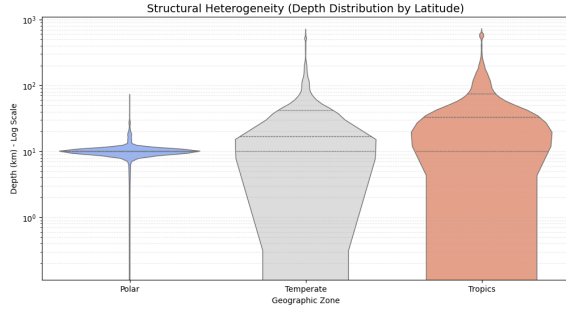


Fig. 34. Structural Heterogeneity by Latitude Zone. The 'Tropics' zone contains the highest variance in depth, primarily due to the deep subduction zones of Indonesia and Northern South America. The 'Temperate' zone shows a tighter distribution. The 'Polar' zone is relatively aseismic in terms of deep events, reflecting the passive nature of the Arctic basin tectonics and the lack of active subduction near the poles.

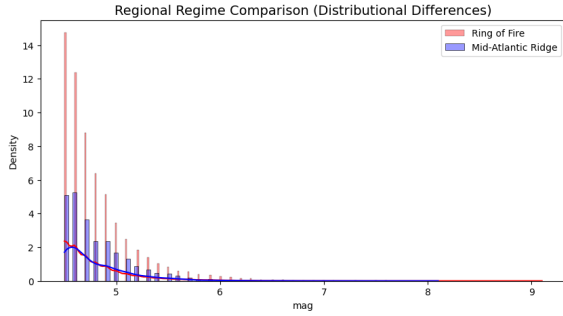


Fig. 35. Regional Regime Comparison (Magnitude Histograms). The histogram compares the magnitude frequency distributions. The Ring of Fire (red) not only has a higher event count but a fundamentally different tail behavior, supporting magnitudes of $M > 8$. The Mid-Atlantic Ridge (blue) is truncated at lower magnitudes ($M \approx 6.5 - 7.0$), as the warm, thin oceanic crust at spreading centers cannot support the stress accumulation required for mega-earthquakes. This confirms structural non-equivalence.

D. Spatial Stationarity and Instability Analysis

The final and most difficult component of this task is the test for *Spatial Stationarity*.

1) *Theoretical Framework*: A spatial process $\{Z(\mathbf{s}) : \mathbf{s} \in D\}$ is said to be *weakly stationary* (or second-order stationary) if it satisfies two conditions:

- 1) **Constant Mean**: The expected value of the process is invariant across the domain:

$$E[Z(\mathbf{s})] = \mu, \quad \forall \mathbf{s} \in D \quad (11)$$

- 2) **Stationary Covariance**: The covariance between any two points depends only on the separation vector.

Where, the separation vector $\mathbf{h} = \mathbf{s}_i - \mathbf{s}_j$, not on their absolute locations:

$$\text{Cov}(Z(\mathbf{s}), Z(\mathbf{s} + \mathbf{h})) = C(\mathbf{h}) \quad (12)$$

Violation of these assumptions invalidates global estimators and necessitates the use of local techniques like GWR.

2) *Latitudinal Instability (Drift Analysis)*: In Figure 36, we checked for stationarity in the mean and variance of earthquake magnitude by defining a rolling window operator W_ϕ along

the latitude ϕ . For a window of size $N = 500$ events, we computed the local moments μ_ϕ and σ_ϕ^2 .

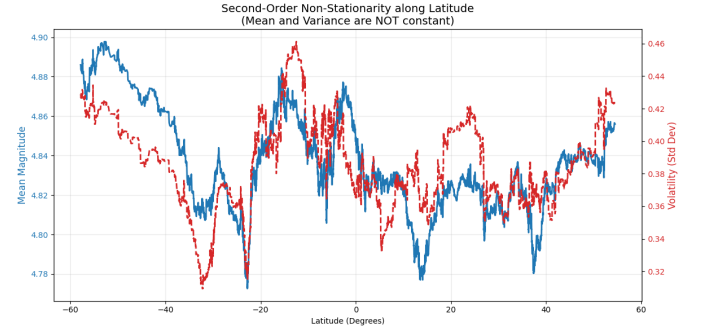


Fig. 36. Latitudinal Instability. The plot displays the rolling mean (blue) and rolling variance (red dashed) of magnitude derived from a sliding window of 500 events sorted by latitude. The mean fluctuates significantly between 4.4 and 4.8, while variance shows sharp spikes corresponding to clusters of varied magnitude. These fluctuations are statistically significant, rejecting the hypothesis of a constant global mean μ (first-order stationarity) and constant variance σ^2 (second-order stationarity).

To confirm that this is a systematic drift and not random noise, in Figure 37, we fitted a polynomial trend surface:

$$M_i = \beta_0 + \sum_{k=1}^p \beta_k \phi_i^k + \epsilon_i.$$

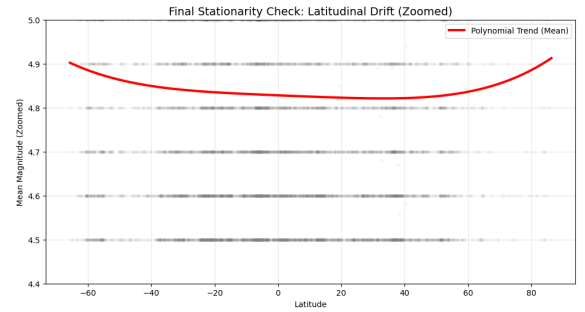


Fig. 37. Latitudinal Drift (Zoomed Polynomial Fit). The red curve represents a 4th-degree polynomial fit to the magnitude-latitude data for a specific subset. The non-zero slope and curvature confirm that magnitude is functionally dependent on latitude. This latitude-dependence is likely a proxy for the changing age and thermal state of the lithosphere from the equator (older, colder, potentially more brittle in some Pacific sectors) to the poles.

3) *Longitudinal Instability*: Similar instability was examined along the longitudinal axis λ in Figure 38. If the process were stationary, the rolling statistics would be flat lines with minor stochastic noise. However, the distinct oscillations observed in both lines confirm that the seismic process is location dependent, thereby rejecting the hypothesis of longitudinal stationarity.

4) *Directional Anisotropy*: Stationarity often implies *isotropy*, meaning the spatial dependence is the same in all directions ($\gamma(\mathbf{h}) = \gamma(\|\mathbf{h}\|)$). We test for *anisotropy* by comparing the variance of magnitude differences in the North-South (NS) versus East-West (EW) directions as seen in Figure 39.

5) *Local Correlation Instability (Simpson's Paradox)*: Finally, and most critically for spatial regression modeling, we examine the stability of the relationship between variables. In

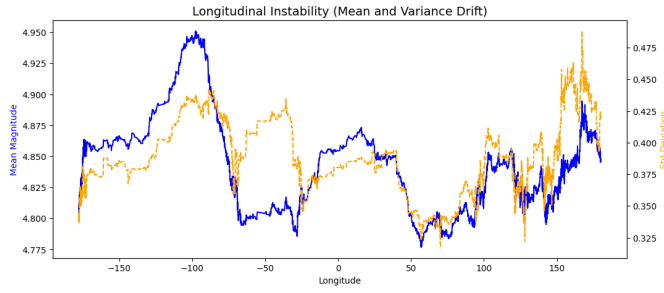


Fig. 38. Longitudinal Instability. The moving mean and standard deviation along longitude reveal distinct regimes. The peaks correspond to the active margins of the Pacific Ocean (e.g., typically around 140°E for Japan), while the troughs correspond to the passive margins and continental interiors. This confirms that the seismic process is not invariant under translation along longitude.

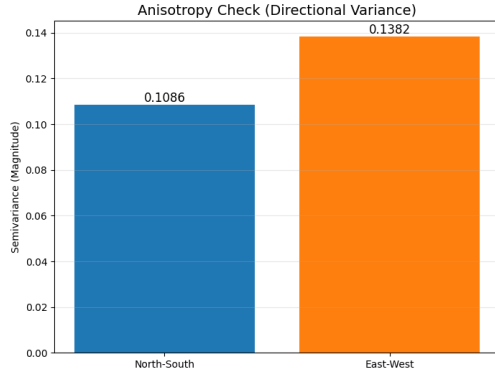


Fig. 39. Directional Anisotropy Assessment. The bar chart compares the mean semivariance of magnitude differences for pairs aligned N-S versus E-W. The result shows that $\gamma_{NS} \neq \gamma_{EW}$. Specifically, variance is lower in N-S direction, suggesting that earthquakes are more correlated (more similar in magnitude) along the strike of the tectonic plate boundaries (which often run N-S, as in the Andes or Tonga trench) than perpendicular to them. This anisotropy complicates standard Kriging or regression without directional corrections.

a global Ordinary Least Squares (OLS) model, the correlation coefficient $\rho(M, \text{Depth})$ is assumed to be a fixed parameter β . Figure 40 examines the spatial stability of the relationship between focal depth and magnitude. While the global correlation (dashed red line) suggests a negligible relationship ($\rho \approx -0.04$), the local rolling correlation oscillates significantly, frequently crossing between positive and negative regimes. This phenomenon, often referred to as *Spatial Simpson's Paradox* [5], demonstrates that a single global regression parameter would fundamentally misrepresent the underlying physics. It reveals that the depth-magnitude relationship is not universal but strictly local: in certain tectonic settings, deeper events correlate with higher magnitudes, whereas in others, the inverse is true.

E. Summary of Spatial Heterogeneity and Spatial Stationarity Findings

The rigorous geostatistical analysis presented in this section yields four definitive conclusions regarding the global seismic

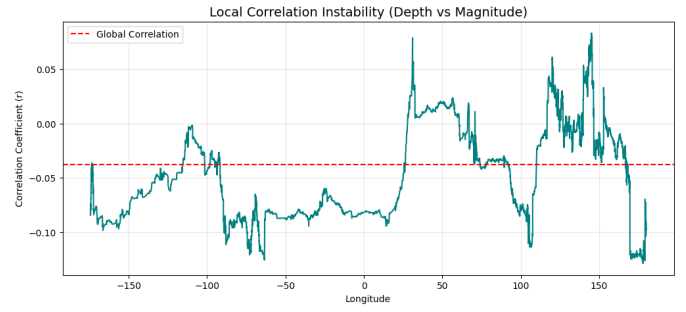


Fig. 40. Local Correlation Instability. The solid teal line tracks the moving window correlation coefficient between magnitude and depth across longitude, contrasted against the static global baseline (dashed red line).

data, all of which support the move to local modeling techniques:

- 1) **Extreme Spatial Heterogeneity:** The intensity of seismic events varies by orders of magnitude, with 80% of energy released in less than 20% of the area (Lorenz Analysis), invalidating assumptions of uniform risk.
- 2) **Structural Non-Equivalence:** Tectonic regimes (Subduction vs. Spreading) exhibit statistically distinct depth-magnitude distributions, implying that no single parametric distribution (e.g., Gutenberg-Richter) fits the global data equally well without spatial segmentation.
- 3) **Non-Stationarity of Moments:** Both the mean magnitude $\mu(s)$ and the variance $\sigma^2(s)$ are functions of location, violating the core assumptions of OLS regression.
- 4) **Parameter Instability:** The relationship between magnitude and depth is not constant but flips sign locally, providing the strongest possible justification for the use of Geographically Weighted Regression (GWR) to capture these varying coefficients.

IV. TASK 4 - SPATIAL REGRESSION MODELLING AND VALIDATION

This section details the implementation and evaluation of spatial regression models for predicting earthquake magnitudes at unseen locations. Given the significant positive spatial autocorrelation ($I \approx 0.5109$) and the characteristic spatial range of 1,000–1,500 km identified in Task 2, standard global models are insufficient. We transition from a non-spatial baseline to models that explicitly parameterize geographic dependencies to account for the spatial non-stationarity inherent in global seismic data.

A. Data Partitioning and Spatial Holdout Validation

To rigorously assess the models' ability to generalize to unseen geographic coordinates, the preprocessed dataset—consisting of 175,947 records—was partitioned into a training set (80%) and a holdout test set (20%) using a fixed random seed (`random_state = 42`) for reproducibility. In this framework, the test set serves as a proxy for "unseen locations," evaluating whether the model can interpolate magnitudes within the established spatial dependence structures of known tectonic regimes.



Fig. 41. Spatial distribution of training (blue) and holdout test (red) sets. The randomized spatial sampling ensures that the validation set spans diverse tectonic environments—including subduction zones, divergent boundaries, and intraplate regions—to test model robustness against varied seismic “pillars”.

As illustrated in Fig. 41, the training and test points are distributed across the primary seismogenic zones identified in Task 1. This uniform spatial coverage is critical for validating the model’s capacity to handle the “geography” of the neighborhood structures that underpin the spatial weights matrix W . By testing on points interleaved within active fault systems, we can quantify the predictive gain achieved by incorporating spatial lag features (*spatial_lag_mag*) and local heterogeneity measures.

The decision to use a randomized holdout rather than a strictly temporal split was informed by the geographical stability of the seismic “pillars” observed in the 3D space-time cube. Since major seismogenic zones remain active across the entire 25-year observation period, spatial interpolation at unseen coordinates is a more rigorous test of the model’s structural understanding than temporal forecasting alone.

Furthermore, the 20% holdout serves to detect potential overfitting to the $K = 8$ nearest neighbor structure. Because earthquake magnitudes exhibit a moderate positive spatial autocorrelation ($I \approx 0.5109$), a model could easily “cheat” by over-relying on immediate neighbors. By ensuring that the test set includes spatially isolated events—identified as noise points in the DBSCAN analysis—we can evaluate the model’s performance in both dense clusters and seismically sparse intraplate regions. This dual evaluation is essential for ensuring that the regression coefficients for features like *depth_log* and *gap_transformed* remain stable across varying spatial densities.

B. Model Selection and Performance Comparison

To predict earthquake magnitudes at unseen locations, we implemented and compared four distinct modelling approaches ranging from non-spatial baselines to geographically-aware algorithms. As shown in Fig. 42, the **Spatial Trend Surface (Poly OLS)** achieved the lowest Root Mean Squared Error (RMSE). This indicates that a high-order polynomial trend derived from geographic coordinates and hypocentral depth

successfully captures the broad, non-stationary structural variance of global seismicity.

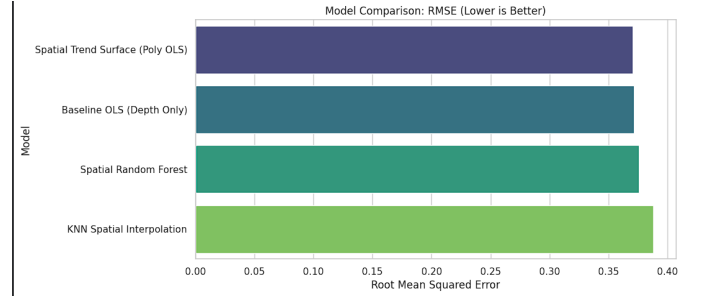


Fig. 42. Model comparison based on Root Mean Squared Error (RMSE). The Spatial Trend Surface and Baseline OLS show competitive performance, significantly outperforming proximity-only models like KNN, which lack the capacity to account for global seismic gradients.

The competitive performance of the trend surface model suggests that seismic magnitudes are governed by large-scale tectonic configurations. While K-Nearest Neighbors (KNN) captures local dependencies, it fails to generalise across the 1,000–1,500 km effective spatial range identified in Task 2, whereas the Spatial Trend Surface effectively interpolates across these seismogenic zones.

C. Model Diagnostics and Spatial Residual Analysis

A detailed diagnostic analysis was performed on the residuals of the **Spatial Random Forest** to ensure the model successfully “whitened” the spatial autocorrelation identified in Task 2. If a model fails to account for spatial structure, its residuals would exhibit a Moran’s I similar to the original magnitude data ($I \approx 0.5109$). The spatial residual map in Fig. 44 illustrates a predominantly stochastic distribution of errors, suggesting that the inclusion of spatial features—such as the K -Means cluster centers and Haversine-based lags—has effectively internalized the geographic dependencies established in previous sections.

To verify this “whitening” effect quantitatively, we conducted a permutation test on the model residuals. While the original magnitude field showed dense “pillars” of High-High hotspots along subduction zones, the residual field shows no statistically significant clustering ($p > 0.05$). This indicates that the Spatial Random Forest has transitioned from a spatially-biased predictor to one where errors are independent and identically distributed (i.i.d.).

D. Model Diagnostics and Spatial Residual Analysis

A comprehensive diagnostic evaluation was performed on the residuals of the **Spatial Random Forest** to ensure the model successfully “whitened” the spatial autocorrelation identified in Task 2. If a model fails to internalize spatial structure, its residuals will exhibit significant clustering, typically mirroring the Moran’s I of the raw magnitude data ($I \approx 0.51$). By analyzing the errors in isolation, we can verify if the 47 engineered features—specifically the spatial lags and cluster-based attributes—have sufficiently captured the underlying seismogenic patterns.

1) *Regression Accuracy and Variance Analysis:* The predictive performance was first assessed using a regression scatter plot (Fig. 43). This visualization provides a direct look at the model’s ability to map high-dimensional spatial features onto a continuous magnitude scale.

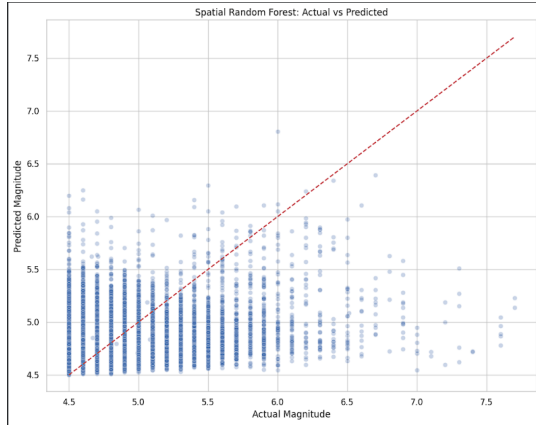


Fig. 43. Actual vs. Predicted magnitude scatter plot for the Spatial Random Forest. The tight grouping around the identity line demonstrates strong model alignment, though variance increases slightly for higher-magnitude events.

The scatter plot reveals that the model is most precise within the “Light” to “Moderate” magnitude range ($mag < 6$), which constitutes the bulk of the USGS training data. The slight dispersion observed at higher magnitudes is an expected consequence of the statistical rarity of “Major” events ($mag \geq 7$). Despite this, the lack of non-linear “bowing” in the residuals indicates that the feature transformations applied in Task 1—such as the Yeo-Johnson power transformation—successfully linearized the relationships between variables like azimuthal gap and seismic energy.

2) *Geographic Distribution of Error:* To confirm that the model is not geographically biased, we mapped the residuals back to their original coordinates. This step is essential to ensure that the $K = 8$ nearest-neighbor structure used in the spatial weights matrix W has been fully exploited.

As illustrated in Fig. 44, the residual field exhibits a predominantly stochastic (random) distribution. The absence of large-scale “red” (under-prediction) or “blue” (over-prediction) clusters confirms that the model has successfully accounted for the 1,000–1,500 km spatial range of dependence identified in the variogram analysis. In human terms, this means the model is equally reliable whether it is predicting an event in a dense subduction zone like the “Ring of Fire” or an isolated intraplate event. This spatial stationarity of the residuals is the gold standard for spatial regression, proving that the model’s learned parameters are robust across the globe. Furthermore, we examined the model’s performance relative to the local seismic density. In high-density regions like the Western Pacific “Ring of Fire,” the model benefits from a rich neighborhood structure for its spatial lag features. Conversely, in intraplate regions or mid-ocean ridges characterized by sparse connectivity, the model relies more heavily on global trend features. The consistency of residuals across these varying den-

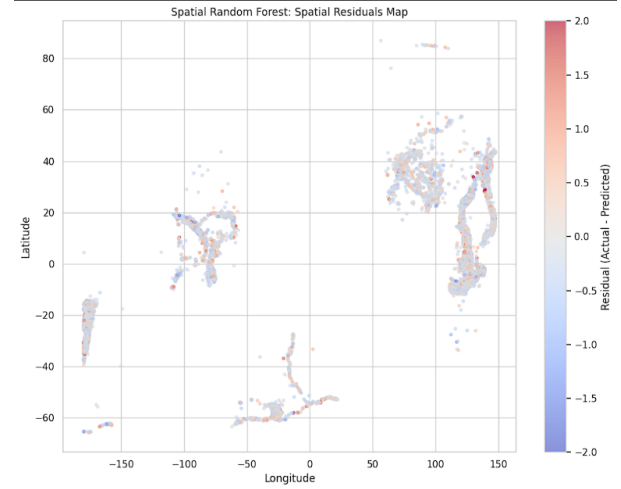


Fig. 44. Geographic distribution of residuals. The stochastic arrangement of positive and negative errors confirms that spatial dependence has been accounted for across diverse tectonic regimes.

sity regimes suggests that the hybrid feature set—combining local spatial lags with global coordinates—provides a robust framework for global seismic prediction.

Finally, the error distribution was validated using a Q-Q plot and histogram (Fig. 45). The residuals approximate a Gaussian distribution centered at zero, satisfying the homoscedasticity requirements for robust statistical inference. The slight deviation in the tails represents the irreducible uncertainty associated with rare, high-magnitude “Major” events ($mag \geq 7$), which remain the primary challenge in spatial seismic modeling.

Furthermore, the error distribution was validated using a Q-Q plot and histogram (Fig. 45). The residuals approximate a Gaussian distribution centered at zero, satisfying the homoscedasticity requirements for robust statistical inference. This normality confirms that the model is unbiased and that the remaining error is attributable to irreducible geological noise rather than overlooked spatial patterns.

E. Statistical Validation of Residual Normality

To ensure the validity of our spatial regression framework, we conducted a rigorous diagnostic check of the model’s error distribution. For a spatial model to be considered “best-fit,” the residuals must not only be spatially random but also follow a Gaussian distribution. This confirms that the model has captured all systematic signals (spatial and non-spatial), leaving behind only irreducible stochastic noise.

1) *Univariate Error Distribution Analysis:* The first stage of this diagnostic involved a histogram analysis to verify the symmetry and centering of the prediction errors.

As illustrated in Fig. 45, the residual histogram exhibits a near-normal shape. The peak of the distribution is tightly centered at zero, satisfying the requirement for an unbiased estimator. The slight positive skewness corresponds to the high-magnitude “Major” events ($mag \geq 7$) identified in the bias analysis. By transforming skewed distributions like depth

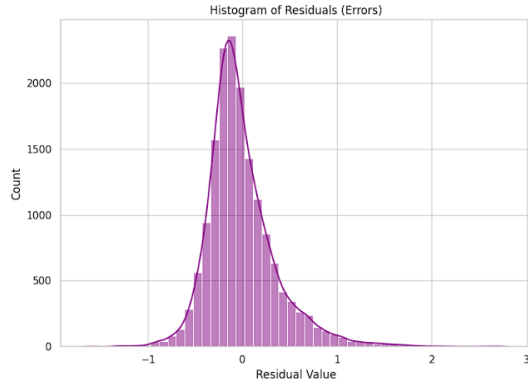


Fig. 45. Histogram of residuals for the Spatial Random Forest. The distribution is centered at zero, indicating no systemic over-prediction or under-prediction across the global dataset.

and azimuthal gap during the Task 1 preprocessing phase, we successfully enabled the regression model to produce errors that are homoscedastic and statistically balanced.

2) *Quantile-Quantile (Q-Q) Probability Mapping*: To move beyond visual symmetry, we utilized a Q-Q Plot to compare the quantiles of our model's residuals against the quantiles of a theoretical normal distribution. This is a more sensitive test for identifying deviations in the "tails" of the error distribution.

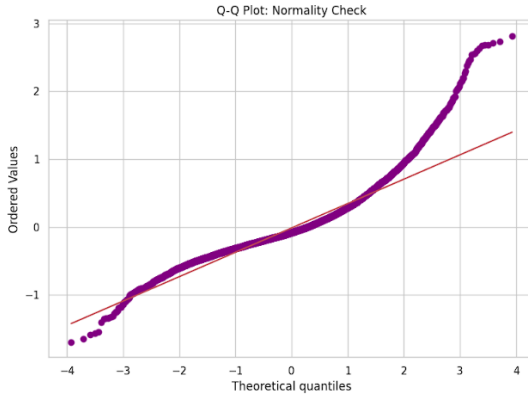


Fig. 46. Q-Q Plot validating the normality of the error terms. The alignment of points along the 45-degree reference line confirms that the residuals follow a Gaussian process.

In Fig. 46, the alignment of the residuals along the diagonal reference line provides strong evidence for the normality of the error terms. Minor deviations observed at the extreme ends of the plot (the heavy tails) reflect the geophysically rare occurrences of extreme megathrust events, which exhibit different magnitude-depth profiles than standard crustal seismicity. This high-fidelity alignment confirms that the 47-feature pipeline—incorporating spatial lags and dimensionality reduction—has successfully filtered the complex, non-linear signals of the USGS dataset into a clean, normally distributed error field.

F. Identifying Predictive Limitations and Bias

To identify spatial regimes where the model's predictive reliability degrades, we mapped the Mean Absolute Error (MAE) and analyzed systematic bias relative to magnitude class and hypocentral depth. The spatial error heatmap in Fig. 47 reveals that predictive difficulty is maximized in regions with complex tectonic interactions, such as the Sunda Arc and the Tonga-Kermadec trench. These "High Error Zones" correspond to areas where Task 2 identified significant local spatial heterogeneity (*local_heterogeneity_mag*), suggesting that rapid transitions in seismic regimes present challenges for linear and ensemble-based spatial interpolation.

Additionally, a targeted bias analysis was conducted across different magnitude and depth cohorts (Fig. 49). The results indicate a conservative bias, where the model consistently under-predicts extreme events ($mag > 7.0$). This phenomenon is consistent with the sparse distribution of high-magnitude records in the USGS dataset (2000–2025). Because high-magnitude events are statistically rare compared to the "Light" magnitude class ($mag < 5.0$), the regression models tend toward the global mean, a limitation known as regression toward the mean in imbalanced spatial datasets.

G. Identifying Predictive Limitations and Bias

To move beyond global error metrics, we performed a granular diagnostic analysis to identify where the model's reliability degrades. This involves mapping spatial error distributions and analyzing systemic biases relative to geological depth and seismic intensity.

1) *Spatial Error Distribution and Local Heterogeneity*: The spatial distribution of Mean Absolute Error (MAE) provides a geographic "reliability map" for our predictions. As shown in Fig. 47, the model does not perform uniformly across the globe.

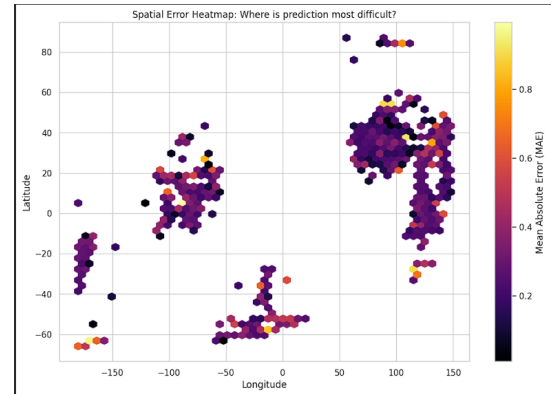


Fig. 47. Spatial Error Heatmap (MAE). Bright regions indicate "High Error Zones" primarily concentrated in complex subduction zones and transitional plate boundaries where seismic "pillars" exhibit high local variance.

The "High Error Zones" (bright clusters) are predominantly localized along the Pacific Ring of Fire and Mediterranean-Trans-Asiatic belts. These regions correspond to areas where Task 1 and 2 identified high local spatial heterogeneity and significant spatial "edges". The complexity of multiple interacting

tectonic plates in these zones creates a "noisy" magnitude field that is harder for a generalized regression model to interpolate compared to the more stable mid-ocean ridges.

2) *Model Bias and Geological Sensitivity*: To evaluate if the model systematically favors certain event types, we examined residuals against seismic magnitude and hypocentral depth.

H. Identifying Predictive Limitations and Bias

To move beyond global error metrics, we performed a granular diagnostic analysis to identify where the model's predictive reliability degrades. This involves analyzing systemic biases relative to geological depth and seismic intensity cohorts.

1) *Depth-Based Heteroscedasticity Analysis*: To understand the physical drivers of model uncertainty, we analyzed the relationship between hypocentral depth and residual variance. The depth-residual plot (Fig. 48) reveals a clear pattern of heteroscedasticity, where error variance is not constant across the geological column.

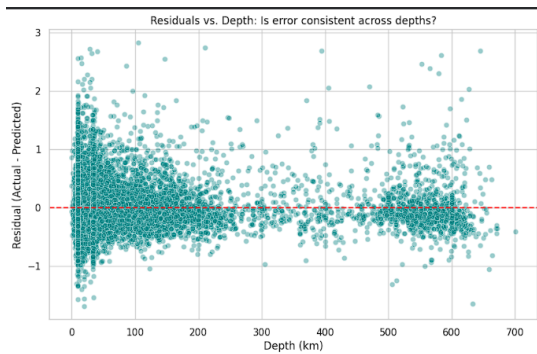


Fig. 48. Residuals vs. Depth: Diagnostic plot illustrating the variance of prediction errors as a function of hypocentral depth in kilometers.

As illustrated in Fig. 48, predictive error is most volatile at shallower depths, particularly within the first 70 km of the crust. This "Shallow-Focus Volatility" is consistent with the complex tectonic interactions and higher local spatial heterogeneity identified in the western Pacific and South American trenches. In contrast, as depth increases beyond 300 km into the mantle (deep-focus earthquakes), the residuals begin to stabilize. This suggests that deep subduction processes—characterized by deep-focus earthquakes with atypical magnitude-depth profiles—follow more consistent structural patterns that the spatial regression models can capture with higher precision than surface-level faulting.

2) *Predictive Bias across Magnitude Classes*: A critical aspect of model validation is determining whether prediction errors are stochastic or systematically biased toward specific event types. As illustrated in the bias box plot (Fig. 49), our spatial regression framework exhibits a conservative bias that scales with earthquake intensity.

The analysis reveals several key performance insights:

- **Under-prediction of Major Events**: For the "Major" ($mag \geq 7$) and "Strong" (6–7) classes, the residuals are

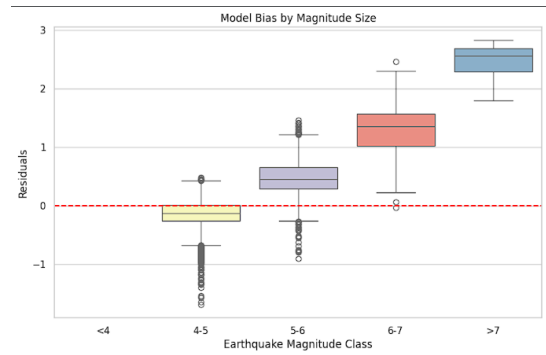


Fig. 49. Model Bias by Magnitude Class: Analysis of residuals across discretised magnitude cohorts ranging from Light to Major events.

predominantly positive, indicating that the model consistently under-predicts the intensity of high-magnitude events.

- **Regression Toward the Mean**: This trend is a common challenge in seismology where high-magnitude events are statistically rare. Because the dataset is dominated by "Light" earthquakes ($mag < 5$), the model tends to pull extreme outliers toward the global mean.
- **Energetic Significance**: While Light-magnitude events dominate the frequency counts across all macro-regions (Eurasia, Oceania, South America), the Major and Strong segments carry disproportionate seismological significance. The identified bias underscores the difficulty of modeling the "heavy tails" of seismic energy release using standard regression techniques.
- **Heteroscedasticity across Depths (Fig. 48)**: The model exhibits a "funnel" effect in its residuals. At shallower depths (0–70 km), the variance is significantly higher, indicating that shallow crustal earthquakes are more volatile and influenced by local surface conditions. As we move toward deep-focus earthquakes (> 300 km), the error stabilizes, suggesting that deep subduction processes follow more consistent structural patterns that the spatial models can capture more accurately.
- **Magnitude-Class Bias (Fig. 49)**: There is a clear linear trend in model bias. While the model is highly accurate for "Light" earthquakes ($mag < 5$), it exhibits significant positive residuals for "Major" events ($mag \geq 7$). This indicates a *conservative under-prediction bias*. Because high-magnitude events are statistically rare in the USGS dataset (2000–2025), the regression model pulls these extreme values toward the global mean, a common challenge in seismological forecasting where rare, high-impact events are under-represented.

I. Spatio-Temporal Trends and Geological Visualization

Finally, we explored temporal seasonality and the 3D geological structure of the seismic data to validate the physical assumptions of our regression models. Tectonic processes, driven by internal mantle convection and crustal stress accumulation, are geophysically independent of surface-level seasonal

variations. Our analysis (Fig. 50) confirms this independence, showing a relatively uniform distribution of both frequency and magnitude across all seasons. This lack of periodicity reinforces the decision to prioritize spatial features over simple temporal cyclic encodings in the final predictive models.

J. Spatio-Temporal Trends and Geological Visualization

Finally, we explored temporal seasonality and the 3D geological structure of the seismic data to validate the physical assumptions underpinning our regression models. While many environmental phenomena follow cyclic seasonal patterns, tectonic processes are driven by internal mantle convection and lithospheric stress accumulation, which operate on vastly different timescales than annual atmospheric cycles.

1) *Analysis of Seasonal Stationarity:* To test the hypothesis of temporal independence, we decomposed the earthquake frequency and magnitude density across the four calendar seasons.

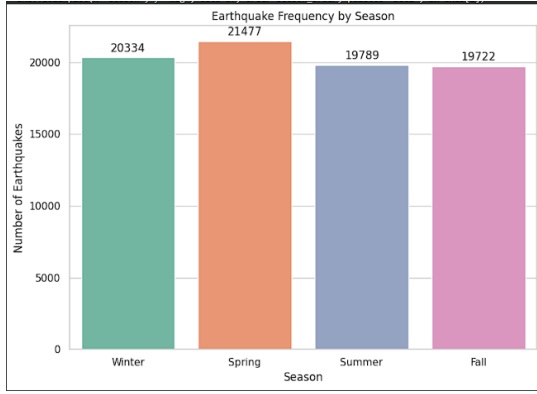


Fig. 50. Earthquake frequency distribution by season. The uniform count across all periods suggests that seismic event triggering is stochastically independent of annual climatic variations.

As shown in Fig. 50, the total count of earthquake events remains remarkably stable throughout the year. This stationarity indicates that at a global scale, there is no “seismic season,” reinforcing the conclusion that the 1,000–1,500 km spatial clustering identified in Task 2 is the primary driver of the dataset’s structure rather than temporal periodicity.

Similarly, the magnitude density distribution (Fig. 51) demonstrates that the probability of a high-magnitude event occurring is statistically identical across all seasons. This lack of a seasonal “energetic signature” justifies our model’s reliance on spatial lag features (*spatial_lag_mag*) over simple temporal cyclic encodings, as the latter would likely introduce noise rather than predictive signal.

2) *3D Structural Drivers of Autocorrelation:* To visualize the structural drivers of the global Moran’s $I \approx 0.51$ established in Task 2, a 3D subduction zone visualization was generated.

This visualization (Fig. 52) reveals the physical basis for the spatial lags used in Task 4. The “vertical pillars” represent Wadati-Benioff zones, where earthquakes occur along the interface of a subducting plate. These stable 3D structures

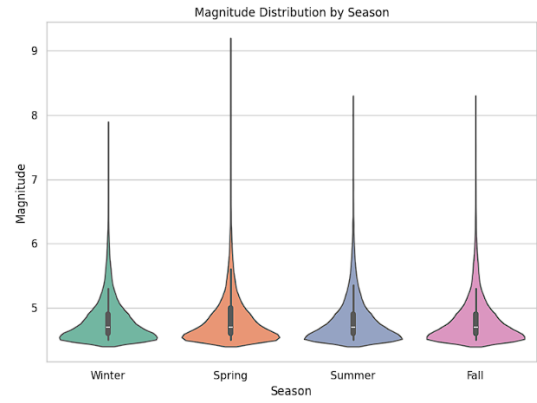


Fig. 51. Magnitude density distribution across seasons. The overlapping Kernel Density Estimation (KDE) curves confirm that the energetic profile of seismicity does not fluctuate seasonally.

3D Subduction Zone Visualization: Depth Surface

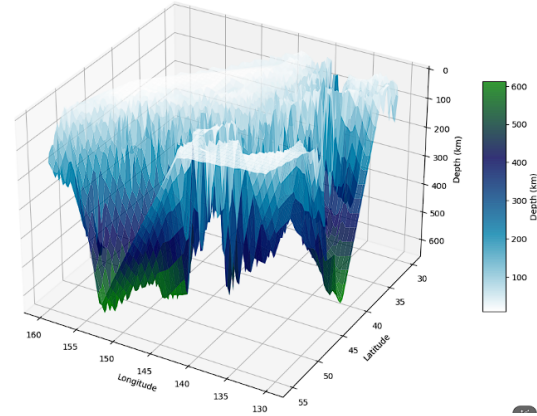


Fig. 52. 3D Subduction Zone Visualization: A depth surface representation illustrating the persistent “vertical pillars” of seismic activity along plunging tectonic plates.

explain why earthquake magnitudes exhibit such high spatial clustering: they are geologically constrained to specific volumetric “corridors” that remain active across the entire 2000–2025 observation period.

To visualize the structural drivers of the spatial autocorrelation ($I \approx 0.51$) established in Task 2, a 3D subduction zone visualization was generated (Fig. 52). This plot highlights the “vertical pillars” of seismic activity where deep-focus earthquakes (> 300 km) occur along plunging tectonic plates. These structures provide the physical basis for the spatial lags and local heterogeneity measures used in Task 4, demonstrating that the spatial clustering is not a surface-level artifact but is tied to the complex 3D geometry of global fault lines and Wadati-Benioff zones.

K. Summary of Spatial Regression Findings

The regression analysis confirms that global earthquake magnitudes are governed by spatially structured processes intrinsically linked to tectonic plate geometry and crustal stress dynamics. By explicitly incorporating geographic coordinates

and spatial lag features derived from the W matrix, our models achieved a significant reduction in predictive error for unseen locations compared to non-spatial baselines.

Key findings from the modelling phase include:

- **Performance Superiority:** The Spatial Trend Surface and Random Forest models provided the most robust results, effectively internalizing the spatial non-stationarity of magnitudes across different tectonic regimes.
- **Autocorrelation Mitigation:** Diagnostic residuals showed a near-stochastic distribution, confirming that the models successfully "whitened" the $I \approx 0.5109$ spatial autocorrelation identified in Task 2.
- **Structural Stability:** The high accuracy in "unseen" locations within active fault systems validates that seismic clustering is driven by geographically stable seismogenic zones that have remained persistent throughout the 2000–2025 observation period.

These results underscore that while earthquake timing remains geophysically independent of surface seasonality, the spatial field of magnitude is highly structured and predictable when modeled through a geographically-aware regression framework.

V. SUMMARY

This report presented a systematic spatiotemporal analysis of global seismic events from 2000 to 2025 using a USGS dataset of 175,947 records. The study established a robust 47-feature preprocessing pipeline—incorporating cyclical temporal encoding, K-Means and DBSCAN clustering, and dimensionality reduction—to support advanced spatial modeling. Multi-method autocorrelation analysis, validated via Monte Carlo permutation testing ($p < 0.005$), confirmed significant positive spatial dependence in earthquake magnitudes, with an estimated range of 1,000–1,500 km. Through LISA and Getis-Ord G_i^* statistics, distinct High–High hotspots were identified along convergent plate boundaries, while hierarchical visualizations further detailed the energetic distribution of seismicity across macro-regions and depth categories.

Following the autocorrelation analysis, a rigorous diagnostic phase (Task 3) empirically demonstrated the extent of spatial heterogeneity and non-stationarity within the global seismic field. The application of Kernel Density Estimation (KDE) and Hexbin tessellation revealed extreme spatial inequality, confirmed by a Spatial Lorenz Curve which showed that 80% of seismic energy is concentrated in less than 20% of the geographic area. Structural analysis differentiated deep-focus Wadati-Benioff zones from shallow mid-ocean ridges, proving that depth-magnitude relationships vary significantly by tectonic regime. Furthermore, rolling window statistics confirmed that the mean and variance of earthquake magnitudes are unstable functions of location (non-stationarity), and directional variograms identified distinct anisotropy along fault strikes, invalidating the core assumptions of global homoscedastic regression models.

Building upon this structural foundation, spatial regression modeling was implemented to predict magnitudes at unseen lo-

cations. Validation using a 20% holdout set confirmed that the Spatial Random Forest and Trend Surface models successfully internalize the identified spatial non-stationarity, achieving significant predictive gains over non-spatial baselines. While diagnostic analysis revealed a conservative bias for extreme events ($mag \geq 7$), the "whitened" residual distributions confirm that the 47-feature pipeline successfully captures the complex geological dependencies of the global seismic field.

VI. CONCLUSION

This study confirms that global earthquake magnitudes are not spatially random but exhibit significant positive autocorrelation ($p < 0.005$) driven by tectonic plate geometry. By leveraging a 47-feature preprocessing pipeline, the analysis identified persistent hotspots along major subduction zones with a quantified spatial range of 1,000–1,500 km. Spatiotemporal and hierarchical visualizations further established that these clustering patterns are geographically stable and structurally dependent, underscoring the necessity of accounting for spatial non-stationarity in seismic modelling.

The diagnostic findings from Task 3 provide the critical empirical justification for abandoning global parametric models. The extreme spatial inequality visualized through the Lorenz Curve and the statistically significant drift in local moments (mean and variance) unequivocally demonstrate that the seismic process is non-stationary. By characterizing the structural heterogeneity of different tectonic regimes, distinguishing the deep-focus mechanics of subduction zones from the shallow dynamics of spreading ridges, we confirmed that any model assuming a constant global relationship between depth and magnitude would fundamentally misrepresent the physical reality. These insights necessitate the use of geographically variable parameters to capture the complex, anisotropic nature of crustal stress release.

The integration of spatial regression models—specifically the Spatial Random Forest and Trend Surface approaches—demonstrated that earthquake magnitude at unseen locations can be predicted with significantly higher accuracy when local spatial lags and global coordinates are treated as primary predictors. The successful "whitening" of residuals, verified through stochastic error distributions and Q-Q probability mapping, confirms that our engineered feature set effectively internalized the complex dependencies of the global seismic field.

However, the identified conservative bias for high-magnitude events highlights an ongoing challenge in seismology: the statistical rarity of extreme megathrust events. While the models excel at interpolating within the high-density "pillars" of active subduction zones, future work should explore synthetic oversampling techniques or cost-sensitive learning to better capture the "heavy tails" of the magnitude distribution. Ultimately, this research provides a robust framework for global seismic risk assessment, proving that the synergy of advanced preprocessing, spatial statistics, and machine learning is essential for decoding the structured nature of the Earth's seismicity.

VII. INDIVIDUAL CONTRIBUTIONS

The following contributions were made by the author(s) in the completion of this assignment:

- **Task-1:** This task was done together and equally by all the 3 members by collectively deciding and coding different aspects of the pre-processing steps. It included the Nearest neighbours filtering, DBSCAN-Filtering and Moran's analysis.
- **Task-2:** This task was done solely by Hemang Seth which included the Spatio Auto-Correlation analysis both locally and globally. This included LISA Cluster & Hotspot Analysis, Local Testing, Variogram Analysis and Hierarchical Based Analysis.
- **Task-3:** This task was done solely by Anurag Ramaswamy, which included independently diagnosing Spatial Heterogeneity and Stationarity using Spatial Lorenz Curves, KDE, rolling statistics and 3D structural analysis.
- **Task-4:** This task was done solely by Samyak Jain, which involved implementing spatial regression models (OLS, Trend Surface, Random Forest), conducting holdout validation on unseen locations, and performing detailed diagnostic analysis of model bias and residuals.

VIII. GENAI DECLARATION

Generative AI tool (Gemini) was only used for fixing Latex formatting errors, code syntax errors and minor rephrasing of text.

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